Commutative Algebra and Algebraic Geometry

Summary by Ari Feiglin (ari.feiglin@gmail.com)

Contents

Note:
Due to the mismatch in concepts taught in the lecture and recitation, some material is mentioned in this summary before it is formally introduced. For example, the spectrum of a ring is discussed in the context of integral algebras, before it is formally defined in its own section. It is advised to read the section on ring spectra early on (before going in depth on commutative algebra).

1 Commutative Algebra

1.1 Rings and Ideals

We recall some basic notions regarding rings.

1.1.1 Definition

A **ring** is a set R equipped with two operations $+$ and $\cdot$, making it into a commutative group and a monoid respectively. Furthermore, they satisfy the distributivity properties:

$$a \cdot (b + c) = ab + ac, \qquad (a + b) \cdot c = ac + bc$$

for all $a, b, c \in R$. The additive and multiplicative identities are denoted 0 and 1 respectively.

Note:
In this course all rings are assumed to be **commutative**: $\cdot$ forms a commutative monoid.

1.1.2 Definition

A **ring morphism** is a function between two rings, $f\colon R \to S$, which preserves structure: $f(x + y) = f(x) + f(y)$, $f(xy) = f(x)f(y)$, and $f(1) = 1$. A **ring isomorphism** is a ring morphism with an inverse (equivalently a bijective ring morphism).

A **subring** of a ring R is a subset $S \subseteq R$ equipped with a ring structure making the inclusion $S \to R$ a ring morphism.

An **ideal** of a ring R is a nonempty subset $I \subseteq R$ closed under addition and multiplication by R: for $r \in R$ and $x \in I$, $rx \in I$. For an ideal $I \subseteq R$, we define its **quotient ring** R/I to be the set of all cosets $[r] = r + I$ with ring structure given by

$$0 = [0] = I, \qquad 1 = [1], \qquad [x] + [y] = [x + y], \qquad [x] \cdot [y] = [xy],$$

equivalently the unique structure making the projection map $\pi\colon R \to R/I, r \mapsto [r]$ a ring morphism.

1.1.3 Proposition

Let $f\colon R \to S$ be a ring morphism, then

(1) For every ideal $J \leq S$, $f^{-1}(J) \leq R$ is an ideal.

(2) In particular f's **kernel**, $\ker f = f^{-1}0$, is an ideal of R. And f's **image**, $\mathrm{im} f = f(R)$ is a subring of S. f then quotients to give a ring isomorphism $R/\ker f \cong \mathrm{im} f$.

(3) For an ideal $I \leq R$, consider the projection $\pi\colon R \to R/I$. Then $J \mapsto \overline{J} = \pi(J)$ and $\overline{J} \mapsto \pi^{-1}\overline{J}$ gives a one-to-one correspondence between ideals of R containing I and ideals of R/I.

1.1.4 Definition

For a ring R and a subset $S \subseteq R$, we denote the smallest ideal containing S by (S). This exists as the arbitrary intersection of ideals is an ideal. When $S = \{x\}$ is a singleton, we denote $(\{x\}) = (x)$; such an ideal is called a **principal ideal**.

1.1.5 Proposition

The ideal generated by S, (S), is the set of all finite R-combinations of elements from S:

$$(S) = \left\{ \sum_{s \in S} r_s s \;\middle|\; r_s \in R, \text{ all but finitely many being zero} \right\}.$$

In particular, $(x) = Rx$ is the set of all multiples of x.

Proof

It is not hard to see that this is an ideal, and clearly every ideal containing S must contain it. ■

1.1.6 Definition

Let R be a ring, an element $x \in R$ is

(1) a **zero divisor** if there is a $y \in R$ such that $xy = 0$;

(2) a **unit** if there is a $y \in R$ such that $xy = 1$;

(3) **nilpotent** if $x^n = 1$ for some $n > 0$.

The set of units of R is denoted $R^\times$ and forms a group.

1.1.7 Definition

A ring $R \neq 0$ is

(1) **reduced** if every $x \neq 0$ is not nilpotent;

(2) an **integral domain** if every $x \neq 0$ is not a zero divisor;

(3) a **field** if every $x \neq 0$ is a unit.

It is easy to see that a field is an integral domain is reduced.

1.1.8 Proposition

For a ring R, the following are equivalent:

(1) R is a field,

(2) R has no non-trivial ideals,

(3) Every ring morphism out of R to a non-zero ring is injective.

Proof

If R is a field, then every nonzero element is a unit and thus the ideal generating it must be all of R. Conversely if R has no non-trivial ideals, then for $x \neq 0$ we must have that $1 \in (x)$ and thus there is a $y \in R$ such that $xy = 1$ so it is a unit.

If R has no non-trivial ideals then for every $f\colon R \to S \neq 0$ we must have $\ker f = 0$, as it is an ideal not containing 1. Thus f is injective. Conversely if $I \neq R$ is an ideal of R, then $\pi\colon R \to R/I$ is injective, but its kernel is I, so $I = 0$. ■

1.1.9 Proposition

If $\{I_\alpha\}_\alpha$ is a collection of ideals from R, then

(1) $\bigcap_\alpha I_\alpha$ is an ideal, the largest ideal contained in each I_α.

(2) $\sum_\alpha I_\alpha = \{\sum_\alpha x_\alpha \mid x_\alpha \in I_\alpha$ all but finitely many are zero$\}$ is an ideal, the smallest ideal containing each I_α.

Proof

The first point is easily verified, and the second one follows because

$$\sum_\alpha I_\alpha = \left(\bigcup_\alpha I_\alpha\right).$$ ■

1.1.10 Definition

If I, J are ideals then we define their **product** to be the ideal generated from all products of elements from I and J:

$$IJ = (\{xy \mid x \in I, y \in J\}).$$

Since I, J are ideals and are thus closed under multiplication from R, for $x \in I$ and $y \in J$ we have that $xy \in I \cap J$. Thus $IJ \subseteq I \cap J$.

1.1.11 Definition

For an ideal $I \subseteq R$, we define its **radical** to be

$$\sqrt{I} = \{r \in R \mid r^n \in I \text{ for some } n \geq 0\}.$$

1.1.12 Lemma

The radical of an ideal is an ideal.

Proof

If $a^n \in I$ and $r \in R$ then $(ra)^n = r^n a^n \in I$ so that $ra \in \sqrt{I}$. And if $a^n, b^m \in I$ then

$$(a+b)^{n+m} = \sum_{i=0}^{n+m} \binom{n+m}{i} a^i b^{n+m-i} \in I,$$

since for each i, if $i < n$ then $n+m-i > m$, so $a+b \in \sqrt{I}$. ■

1.2 Prime and maximal ideals

1.2.1 Definition

A proper ideal $I \subset R$ is

(1) **prime** if $ab \in I$ implies $a \in I$ or $b \in I$. Equivalently if $a, b \notin I$ then $ab \notin I$.

(2) **maximal** if it is not contained in any other proper ideal.

1.2.2 Proposition

An ideal I is

(1) prime iff R/I is an integral domain;

(2) maximal iff R/I is a field.

Consequently, a maximal ideal is prime.

Proof

R/I is an integral domain iff $[a][b] \neq [0]$ when $[a], [b] \neq [0]$. This means that $ab \notin I$ when $a, b \notin I$, as desired. R/I is a field iff it has no nontrivial ideals. Since ideals of R/I are in bijection with ideals containing I, this is iff I is maximal. ■

In particular the zero ideal $0 = (0) = \{0\}$ is prime iff $R \cong R/(0)$ is an integral domain.

1.2.3 Corollary

Let I be an ideal, then the correspondence $J \leftrightarrow \overline{J}$ between ideals containing I and ideals of R/I preserves primality and maximality: J is prime/maximal iff $\overline{J}$ is.

Proof

We have that

$$(R/I)/\overline{J} = (R/I)/(J/I) \cong R/J,$$

so that $(R/I)/\overline{J}$ is an integral domain/field iff R/J is. The result follows. ■

1.2.4 Definition

(1) An element $0 \neq x \in R$ is **irreducible** if it is not a unit, and if $x = yz$ then either y or z is a unit.

(2) An integral domain R is a **principal ideal domain** (or a **PID**) if all of its ideals are principal.

1.2.5 Proposition

If R is a PID and $0 \neq x$, then (x) is prime iff x is irreducible iff (x) is maximal.

Proof

If (x) is prime, x is not a unit since it is a proper ideal. And if $x = yz$ we have that y or z is in (x), say $y = xt$. Then $x = yz = xtz$, and so $x(1 - tz) = 0$, and since R is an integral domain this means $tz = 1$ so z is a unit.

Now if x is irreducible, then suppose $(x) \subseteq I$ is a proper ideal containing (x). Since R is a PID, $I = (y)$ so that $x \in (y)$ meaning $x = yz$. (y) is a proper ideal so y is not a unit, thus z is, and then $y = xz^{-1}$ so that $y \in (x)$ and thus $(x) = (y)$. So (x) is maximal, as desired. ∎

1.2.6 Example

The ring of integers $\mathbb{Z}$ is a PID. Its irreducible elements are precisely its prime elements, so that its prime ideals are (0) and (p) for $p \in \mathbb{Z}$ prime.

1.2.7 Example

If k is a field, then its ring of polynomials $k[x]$ is a PID. Indeed, if $0 \neq I \leq k[x]$ is an ideal, then let $f \in I$ have minimal degree. By polynomial division we see that $I = (f)$: for $g \in I$ we have that $g = fq + r$, but since $r = g - fq \in I$ has smaller degree than f, it must be zero.

Its prime ideals are (0) and (f) for an irreducible monic polynomial f.

1.2.8 Lemma

If $f\colon R \to S$ is a ring morphism, and $\mathfrak{q} \leq S$ is prime, then $f^{-1}\mathfrak{q} \leq R$ is prime.

Proof

A direct proof is easy, instead consider the composition:

$$\overline{f}\colon R \xrightarrow{f} S \xrightarrow{\pi} S/\mathfrak{q}.$$

By definition $f^{-1}\mathfrak{q} = \ker \overline{f}$, and so $\overline{f}$ induces an embedding $R/f^{-1}\mathfrak{q} \hookrightarrow S/\mathfrak{q}$. Subrings of integral domains are integral domains, so $R/f^{-1}\mathfrak{q}$ is an integral domain, thus $f^{-1}\mathfrak{q}$ is prime. ∎

Subrings of fields are not fields, so the above proof does not show that the preimage of a maximal ideal is maximal. Indeed, this is not true: consider the unique map $f\colon \mathbb{Z} \to \mathbb{Q}$. Then $0 \leq \mathbb{Q}$ is maximal, but $f^{-1}0 = \ker f = 0$ is not maximal in $\mathbb{Z}$.

1.2.9 Theorem

For every ring R and every proper ideal $I < R$, there is a maximal ideal $\mathfrak{m}$ containing I.

Proof

Consider the set of all proper ideals containing I. This is nonm-empty, as it contains I. Furthermore, the union of every chain of proper ideals is a proper ideal; it being an ideal is easy to see, and it is proper since 1 is not in any element of the chain. Thus by Zorn, this set has a maximal element, which must be a maximal ideal. ■

1.2.10 Corollary

Let $x \in R$, then x is a unit iff x is not in any maximal ideal. That is,

$$R^\times = \bigcap_{\mathfrak{m}} \mathfrak{m}^c,$$

where the intersection is over all maximal ideals of R.

Proof

An element is a unit iff it is not contained in any proper ideal. This is clear enough: if x is not a unit then (x) is proper, and proper ideals don't contain units. Every proper ideal can be extended to a maximal ideal, so the result follows. ■

1.2.11 Definition

Let R be a ring. Then define its **nilradical** to be $\mathrm{nil}(R) = \sqrt{0}$, i.e. the ideal of all nilpotent elements.

Clearly R is reduced iff $\mathrm{nil}(R) = 0$.

1.2.12 Proposition

Let $I \leq R$ be an ideal, and let $\pi\colon R \to R/I$ be the projection. Then $\sqrt{I} = \pi^{-1}\mathrm{nil}(R/I)$.

Proof

We note that if $x^n \in I$ then $\pi(x)^n \in I$ so that $\pi(x) \in \mathrm{nil}(R/I)$ as desired. Conversely, if $\pi(x) \in \mathrm{nil}(R/I)$ then $\pi(x^n) = I$ so that $x^n \in I$ as desired. ■

1.2.13 Proposition

For a ring R,

(1) $R/\mathrm{nil}(R)$ is reduced;

(2) The nilradical is the intersection of all prime ideals:

$$\mathrm{nil}(R) = \bigcap_{\mathfrak{p}} \mathfrak{p}.$$

Proof

We noted above that $\mathrm{nil}R = \sqrt{\mathrm{nil}R} = \pi^{-1}\mathrm{nil}(R/\mathrm{nil}(R))$. But we have also that $\mathrm{nil}R = \pi^{-1}0$, so $\pi^{-1}0 = \pi^{-1}\mathrm{nil}(R/\mathrm{nil}(R))$ and since π is surjective this means $\mathrm{nil}(R/\mathrm{nil}(R)) = 0$ so that $R/\mathrm{nil}(R)$ is reduced.

Clearly if $x \in \mathrm{nil}(R)$ then $x \in \mathfrak{p}$ for every prime $\mathfrak{p}$ since $x^n = 0 \in \mathfrak{p}$. Now if $x \notin \mathrm{nil}(R)$ then we claim there exists a prime ideal $\mathfrak{p}$ such that $x \notin \mathfrak{p}$. We prove this using Zorn. Let $T = \{1, x, x^2, \dots\}$ and let S be the poset of all ideals I such that $I \cap T = \emptyset$; note that all ideals in S are proper. As explained before the union of a chain of ideals is an ideal, and clearly if all of the ideals in the union don't intersect T, neither does the union. Thus by Zorn, S has a maximal element $\mathfrak{p}$.

We claim that $\mathfrak{p}$ is prime. Indeed, suppose that $ab \in \mathfrak{p}$ but $a, b \notin \mathfrak{p}$. Then by maximality we have that $x^n \in (a, \mathfrak{p})$ and $x^m \in (b, \mathfrak{p})$ for some $n, m \geq 0$. But then $x^{n+m} \in (a, \mathfrak{p})(b, \mathfrak{p}) \subseteq \mathfrak{p}$, a contradiction.

Thus there is a prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \cap T = \emptyset$, in particular $x \notin \mathfrak{p}$. So if $x \notin \mathrm{nil}(R)$ then $x \notin \bigcap_{\mathfrak{p}} \mathfrak{p}$, as desired. ∎

1.2.14 Corollary

Let I be an ideal of R, then

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p},$$

where the intersection is over all prime ideals containing I.

Proof

We showed above that $\sqrt{I} = \pi^{-1}\mathrm{nil}(R/I)$. By above $\mathrm{nil}(R/I)$ is the intersection over all prime ideals of R/I, which are all prime ideals of R containing I:

$$\sqrt{I} = \pi^{-1} \bigcap_{I \leq \mathfrak{p}} \overline{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \pi^{-1}\overline{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p}.$$ ∎

1.2.15 Definition

Let R be a ring, then define its **Jacobson radical** to be

$$J(R) = \bigcap_{\mathfrak{m}} \mathfrak{m},$$

the intersection of all maximal ideals.

1.2.16 Lemma

Let $\mathfrak{m} \leq R$ be maximal. Then $x \notin \mathfrak{m}$ iff there is a $y \in R$ such that $1 - xy \in \mathfrak{m}$.

Proof

$x \notin \mathfrak{m}$ iff $\overline{x} \in R/\mathfrak{m}$ is nonzero, or equivalently is a unit. This is iff there is a $y \in R$ such that $\overline{xy} = \overline{1}$. This just means $1 - xy \in \mathfrak{m}$, as desired. ∎

1.2.17 Corollary

For a ring R, its Jacobson radical $J(R)$ is the set of all elements $x \in R$ such that $1 - yx \in R^\times$ for all $y \in R$.

Proof

Now if $x \notin J(R)$ then $x \notin \mathfrak{m}$ for some $\mathfrak{m}$, and then $1 - yx \in \mathfrak{m}$ for some y and thus is not a unit. And if $1 - yx$ is not a unit, then $1 - yx \in \mathfrak{m}$ for some maximal $\mathfrak{m}$ and thus $x \notin \mathfrak{m}$ so that $x \notin J(R)$. ■

1.2.18 Definition

A ring R is **local** if it has a unique maximal ideal $\mathfrak{m}$. The field $k = R/\mathfrak{m}$ is called its **residue field**. By abuse of notation we write $(R, \mathfrak{m})$ for a local field.

1.2.19 Proposition

(1) If $(R, \mathfrak{m})$ is a local ring then $R - \mathfrak{m} \subseteq R^\times$.

(2) Conversely if $I < R$ is a proper ideal with $R - I \subseteq R^\times$ then (R, I) is a local ring.

(3) If $\mathfrak{m} \leq R$ is maximal such that $1 + \mathfrak{m} \subseteq R^\times$ then $(R, \mathfrak{m})$ is local.

Proof

If $x \notin \mathfrak{m}$ then (x) is not contained in $\mathfrak{m}$, and since $\mathfrak{m}$ is the unique maximal ideal this means that $(x) = R$ so that x is a unit. Also we noted that $R^\times$ is the intersection of all complements of maximal ideals, so this follows from that as well.

Conversely, we have that

$$R - I \subseteq R^\times \bigcap_{\mathfrak{m}} \mathfrak{m}^c \implies I \supseteq \bigcup_{\mathfrak{m}} \mathfrak{m},$$

which means that I contains all maximal ideals, which can only happen if it is the unique maximal ideal.

We want to show that $R - \mathfrak{m} \subseteq R^\times$. Indeed, if $x \notin \mathfrak{m}$ then there is a y such that $1 - yx \in \mathfrak{m}$ so that $yx \in 1 + \mathfrak{m}$ and is thus a unit. So x is a unit, as desired. ■

1.2.20 Example

For $p \in \mathbb{Z}$ prime, define

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \;\middle|\; a, b \in \mathbb{Z}, b \notin (p) \right\} \subseteq \mathbb{Q}, \qquad I = \left\{ \frac{a}{b} \in \mathbb{Z}_{(p)} \;\middle|\; a \in (p) \right\}.$$

Then $(\mathbb{Z}_{(p)}, I)$ is local. Indeed, if $\frac{a}{b} \notin I$ then it is invertible (its inverse is $\frac{b}{a}$). We can generalize this, as we will in the next section.

1.3 Localization of rings

Consider the following problem: for a ring A and a subset $S \subseteq A$, we want a ring $S^{-1}A$ and a morphism $\iota\colon A \to S^{-1}A$ such that any morphism $f\colon A \to B$ such that $f(S) \subseteq B^\times$, f factors uniquely through ι. That is, there is a unique $\hat{f}\colon S^{-1}A \to B$ such that $f = \hat{f} \circ \iota$:

$$\begin{tikzcd} A \arrow[r, "\iota"'] \arrow[dr, "f"'] & S^{-1}A \arrow[d, dashed, "\exists!"] \\ & B \end{tikzcd}$$

This is a universal property, and as such any solution is unique up to unique isomorphism. In this section we will investigate the construction of such a solution.

1.3.1 Definition

A subset $S \subseteq A$ is **multiplicatively closed** if $1 \in S$ and for $a, b \in S$ we have $ab \in S$.

It is sufficient to consider only the case that S is multiplicatively closed. Otherwise, let $\hat{S}$ be S's multiplicative closure: the set of all finite products of elements from S. Then $\hat{S}^{-1}R$ is a solution to the problem for S. Indeed, if $f(S) \subseteq B^\times$, then $f(\hat{S}) \subseteq B^\times$ as well, as the product of units. Then there is a unique factoring through $\hat{S}$'s ι.

1.3.2 Definition

For $S \subseteq A$ multiplicatively closed, define an equivalence on $A \times S$ by $(a, s) \sim (b, t)$ iff there is a $u \in S$ such that $u(at - bs) = 0$, in which case we write $(a, s) \sim_u (b, t)$.

1.3.3 Lemma

This is indeed an equivalence relation.

Proof

The subrelations $\sim_u$ are all clearly reflexive and symmetric, thus their union $\sim$ is as well. Now if $(a, s) \sim_\alpha (b, t) \sim_\beta (c, v)$ then $\alpha at = \alpha sb$ and $\beta bv = \beta ct$. So then $\alpha t\beta av = \alpha sb\beta v = \alpha\beta tcs$ so that $(a, s) \sim_{\alpha t\beta} (c, v)$. ∎

1.3.4 Example

A more naive equivalence would be $\sim_1$, i.e. $(a, s) \sim (b, t)$ iff $at - bs = 0$. This corresponds to $\sim$ when A is an integral domain, but is not generally an equivalence.

1.3.5 Definition

For $S \subseteq A$ multiplicatively closed, define $S^{-1}A$ to be the quotient of $A \times S$ by $\sim$. Write the equivalence class of (a, s) by a/s. We give this a ring structure by

$$0 = 0/1, \qquad 1 = 1/1, \qquad a/s + b/t = (at + bs)/st, \qquad a/s \cdot b/t = (ab)/(st).$$

Define $\iota \colon A \to S^{-1}A$ by $\iota(a) = a/1$.

1.3.6 Proposition

$S^{-1}A$ is indeed a ring, and ι is a ring morphism.

Proof

We need to show that addition and multiplication are well-defined. Indeed, if $(a,s) \sim_u (a',s')$ and $(b,t) \sim_v (b',t')$ then $(at+bs, st) \sim_{uv} (a't'+b's', s't')$ which can be verified. And similarly $(ab, st) \sim_{uv} (a'b', s't')$. From well-definedness it is easy to see that addition and multiplication define a ring structure and ι is a ring morphism. ∎

A few properties are immediate:

(1) For $s \in S$, $\iota(s) = 1/s$ has an inverse $s/1$, so that $\iota(S) \subseteq (S^{-1}R)^\times$;

(2) We have that $a/s = 0/1$ iff there is a $t \in S$ such that $at = 0$. In particular, $\iota(a) = 0$ iff there is a $t \in S$ such that $ta = 0$.

(3) Every element of $S^{-1}A$ is of the form

$$a/s = a/1 \cdot 1/s = \iota(a) \cdot \iota(s)^{-1}.$$

1.3.7 Theorem

This construction solves the above universal property.

Proof

Let $f\colon A \to B$ be such that $f(S) \subseteq B^\times$: we need to show that there is a unique map $g\colon S^{-1}A \to B$ such that $f = g \circ \iota$. Indeed, we must define

$$g(a/s) = g(\iota(a)\iota(s)^{-1}) = (g \circ \iota(a)) \cdot (g \circ \iota(s))^{-1} = f(a)f(s)^{-1}.$$

This shows that g is unique, if it exists. We need to show that this map is well-defined and a ring morphism. If $a/s = b/t$ then we must have that $f(a)f(s)^{-1} = f(b)f(t)^{-1}$ so that g is well-defined. Indeed, if $u(at - bv) = 0$ for $u \in S$, we have

$$f(u) \cdot (f(at) - f(bv)) = 0$$

and since $f(S) \subseteq B^\times$ we have that $f(u)$ is a unit so that $f(at) - f(bv)$, meaning this is well-defined. And this is a ring morphism, as can be checked easily. ∎

1.3.8 Corollary

Let $f\colon A \to B$ be a ring morphism such that

(1) $f(S) \subseteq B^\times$,

(2) if $f(a) = 0$ then $ta = 0$ for some $t \in S$,

(3) every element of B is of the form $f(a)f(s)^{-1}$ for $a \in A$ and $s \in S$.

Then there is a unique isomorphism $g\colon S^{-1}A \to B$ such that $g \circ \iota = f$.

Proof

f solves the universal problem above for S as well. If $h\colon A \to C$ is a map where $h(S) \subseteq C^\times$, then we need to define $\hat{h}\colon B \to C$ by
$$\hat{h}(f(a)f(s)^{-1}) = h(a)h(s)^{-1},$$
which is well-defined and a ring morphism for similar reasons as above. Since universal properties are unique up to unique isomorphism, the result follows.

Alternatively we can prove this directly: let $g\colon S^{-1}A \to B$ be the unique map where $g \circ \iota = f$ by the universal property. By definition $g(a/s) = f(a)f(s^{-1})$. By the third point g is surjective, and it is injective since if $g(a/s) = 0$ we have $f(a) = 0$ so that $ta = 0$, which means that $a/s = 0$. ■

1.3.9 Example

If A is an integral domain, then $S = A - 0$ is multiplicatively closed, and $S^{-1}A$ is the field of fractions of A.

1.3.10 Proposition

For $A \neq 0$, then $S^{-1}A = 0$ iff $0 \in S$.

Proof

We have that $S^{-1}A = 0$ iff $1/1 = 0/1$ iff there is a $s \in S$ such that $s \cdot 1 = 0$. ■

1.3.11 Definition

Let $\mathfrak{p} \subseteq A$ be prime. Then $S = A - \mathfrak{p}$ is multiplicatively closed, and we denote its localization by $A_\mathfrak{p} = S^{-1}A$.

For $a \in A$, $S = \{1, a, a^2, \dots\}$ is multiplicatively closed, and we denote its localization by $A_a = S^{-1}A$.

For $a \in A$ not nilpotent (so $A_a \neq 0$) define a map
$$\varphi\colon A[x] \to A_a, \qquad x \mapsto 1/a.$$
This map is clearly surjective, and $p(x) = \sum_k^n b_k x^k \in A[x]$ is mapped to zero iff
$$\sum_{k=0}^n b_k \frac{1}{a^k} = \frac{\sum_k b_k a^{n-k}}{a^n} = 0.$$
This is iff there is a t such that
$$a^t \sum_k b_k a^{n-k} = 0.$$
Defining c_k recursively by $c_0 = -b_0$ and $c_k = ac_{k-1} - b_k$ gives a polynomial such that $p(x) = (ax-1)q(x)$, so $p \in (ax-1)$ (the equality above ensures the recursion terminates). Thus $\ker\varphi = (ax-1)$ and thus
$$A_f \cong \frac{A[x]}{(ax-1)}.$$

Note:

Note that we have two distinct localizations here, which have similar names in some cases. If A is a PID and $p \in A$ is irreducible, then $A_{(p)}$ is localization by $A-(p)$, which A_p is localization by $\{p^n\}_{n<\omega}$. Elements of the former are of the form a/s where p does not divide s, while elements of the former are of the form a/s where $s = p^n$. Quite different!

1.3.12 Definition

For a ring morphism $f\colon A \to B$ and an ideal $I \subseteq A$, define the ideal $IB = (f(I))$ of B. Explicitly, IB is the set of all finite sums $\sum_i f(a_i)b_i$ for $a_i \in I$.

In general we have $f^{-1}(f(I)) \supseteq I$ and $f(f^{-1}(J)) \subseteq J$. These imply $f^{-1}(IB) \supseteq I$ and $f^{-1}(J)B \subseteq J$.

1.3.13 Definition

For two ideals I, J, define $(I : J) = \{x \in A \mid xJ \subseteq I\}$. This is an ideal containing I. For $a \in A$, define $(I : a) = (I : (a)) = \{x \in A \mid xa \subseteq I\}$.
For $S \subseteq R$ multiplicatively closed, define the ideal $S^{-1}I = IS^{-1}A \subseteq S^{-1}A$. This is the ideal generated by $\iota(I)$.

So we have two maps: $S^{-1}\bullet$ mapping ideals of A to ideals of $S^{-1}A$, and ι^{-1} mapping ideals of $S^{-1}A$ to A:

$$\{\text{Ideals of } A\} \underset{\iota^{-1}}{\overset{S^{-1}}{\rightleftarrows}} \left\{\text{Ideals of } S^{-1}A\right\}.$$

We wish to relate the two.

1.3.14 Proposition

(1) For an ideal $I \subseteq A$, $S^{-1}I = \{a/s \mid a \in I, s \in S\}$.

(2) For an ideal $J \subseteq S^{-1}A$, we have $S^{-1}\iota^{-1}J = J$.

(3) For an ideal $I \subseteq A$, we have $\iota^{-1}S^{-1}I = \bigcup_{s\in S}(I : s)$.

(4) We have $S^{-1}I = S^{-1}A$ iff $I \cap S \neq \emptyset$.

(5) For $\mathfrak{p} \subseteq A$ prime, if $I \cap S \neq \emptyset$ then $S^{-1}\mathfrak{p} \subseteq S^{-1}A$ is prime (otherwise it is all of $S^{-1}A$).

(6) The correspondence $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$ is a bijection between prime ideals of $S^{-1}A$ and prime ideals of A not intersecting S. Its inverse is S^{-1}. That is, we have a one-to-one correspondence:

$$\{\text{Prime ideals of } A \text{ not intersecting } S\} \underset{\iota^{-1}}{\overset{S^{-1}}{\rightleftarrows}} \left\{\text{Prime ideals of } S^{-1}A\right\}.$$

Proof

(1) $\{a/s \mid a \in I, s \in S\}$ is an ideal and contains $\iota(I)$. It is clearly closed under multiplication by $S^{-1}A$ and for $a/s + b/t = (at + bs)/(st)$, notice that $at + bs \in I$. By definition $S^{-1}I$ is the smallest ideal containing $\iota(I)$, and it consists of finite sums $\sum_i \iota(a_i)x_i/s_i$. This clearly contains a/s, and thus they must be equal.

(2) As explained before, in general we have $S^{-1}\iota^{-1}J = \iota^{-1}(J)S^{-1}A \subseteq J$, we must prove the opposite inclusion. As above, we have $S^{-1}\iota^{-1}J = \{a/s \mid \iota(a) \in J, s \in S\}$. For $x/s \in J$, we have that $\iota(x) = x/1 \in J$ and so $x/s \in S^{-1}\iota^{-1}J$ as desired.

(3) If $a \in \iota^{-1}S^{-1}I$ then $\iota(a) \in S^{-1}I$ so by the first point $a/1 = b/s$ for $b \in I$. Then $asu = bu$ for some $u \in S$ and since $bu \in J$ this means that $a \in (I : su)$. Conversely if $a \in (I : s)$ for $s \in S$ then $as \in I$ and so $\iota(a) = a/1 = (as)/s \in S^{-1}I$.

(4) If $S^{-1}I = S^{-1}A$ then $1/s = a/t$ for $a \in I$, so $as = t$ meaning $as \in I \cap S$. Conversely if $a \in I \cap S$ then $1/1 = a/a \in S^{-1}I$.

(5) If $S \cap \mathfrak{p} = \emptyset$ then we need to show that $S^{-1}\mathfrak{p}$ is prime. If $a_1/s_1, a_2/s_2 \notin S^{-1}\mathfrak{p}$ we want to show $(a_1a_2)/(s_1s_2) \notin S^{-1}\mathfrak{p}$. By assumption $a_1, a_2 \notin \mathfrak{p}$.

Now we claim that if $a \notin \mathfrak{p}$, $a/s \notin S^{-1}\mathfrak{p}$, which would complete the proof. Assume the contrary: $a/s = b/t$ for $b \in \mathfrak{p}$ then there is a $u \in S$ such that $atu = bsu$. Then $atu \in \mathfrak{p}$, but $a \notin \mathfrak{p}$ and $t, u \in S$ and thus also are not in $\mathfrak{p}$, a contradiction to $\mathfrak{p}$'s primality.

(6) Preimages of prime ideals are prime, so ι^{-1} maps prime ideals to prime ideals. By the second point $S^{-1}\iota^{-1}\mathfrak{q} = \mathfrak{q}$ for all prime ideals $\mathfrak{q} \subseteq S^{-1}A$. It remains to show that $\iota^{-1}S^{-1}\mathfrak{p} = \mathfrak{p}$ for all prime $\mathfrak{p} \subseteq A$ which don't intersect S. We know that $\mathfrak{p} \subseteq \iota^{-1}S^{-1}\mathfrak{p}$ as is true in general, so we want to show that $\iota^{-1}S^{-1}\mathfrak{p} = \bigcup_{s\in S}(\mathfrak{p} : s)$ is contained in $\mathfrak{p}$. Indeed, if $a \in (\mathfrak{p} : s)$ then $as \in \mathfrak{p}$ and since $s \notin \mathfrak{p}$ we have that $a \in \mathfrak{p}$ as desired. ■

1.3.15 Corollary

Let $\mathfrak{p} \subseteq A$ be prime, and consider its localization $A_\mathfrak{p}$. Then $A_\mathfrak{p}$ is a local ring with maximal ideal $\mathfrak{p}A_\mathfrak{p}$. Moreover, the correspondence $\mathfrak{q} \to \mathfrak{q}A_\mathfrak{p}$ is a bijection between prime ideals of $A_\mathfrak{p}$ and prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$.

Proof

Let $S = A - \mathfrak{p}$, so that $A_\mathfrak{p} = S^{-1}A$, and $\mathfrak{p}A_\mathfrak{p} = S^{-1}\mathfrak{p}$. The last point is clear: $\mathfrak{q} \cap S = \emptyset$ iff $\mathfrak{q} \subseteq \mathfrak{p}$. $A_\mathfrak{p}$ being local is also clear: $S^{-1}\bullet$ is order-preserving, so $S^{-1}\mathfrak{q} \subseteq S^{-1}\mathfrak{p}$ for all prime ideals $\mathfrak{q}$. Thus $S^{-1}\mathfrak{p}$ is maximal. ■

1.3.16 Corollary

If $a \in A$ is not nilpotent, then there is a prime ideal $\mathfrak{p} \subseteq A$ such that $a \notin \mathfrak{p}$.

Proof

As before, consider the multiplicative set $S = \{a^n\}_{n<\omega}$, and $A_a = S^{-1}A$. Since a is not nilpotent, $0 \notin S$, so that $A_a \neq 0$. So take a prime ideal $\mathfrak{p} \subseteq A_a$, and then $\iota^{-1}\mathfrak{p}$ is a prime ideal of A not intersecting S, thus not containing a. ■

Note:
This completes the proof before that

$$\mathrm{nil}(A) = \bigcap_{\mathfrak{p}} \mathfrak{p}$$

but without appealing to Zorn, and thus not relying on the Axiom of Choice.

1.4 Modules

1.4.1 Definition

Let A be a ring, then a **module** over A is an Abelian group M along with a ring morphism $\varphi\colon A \to \operatorname{end}(M)$. We define **multiplication** to be a function $\mu\colon A \times M \to M$, written by juxtaposition, by

$$\mu(a, m) = \varphi(a)m.$$

An A-module can also be defined via multiplication: require that multiplication distributes both ways, and $1 \in A$ is an identity:

$$(a+b)m = am + bm, \qquad a(m+n) = am + an, \qquad a(bm) = (ab)m, \qquad 1m = m.$$

1.4.2 Definition

For two A-modules M and N, an **A-module morphism** is a map $f\colon M \to N$ which is an Abelian group morphism and also multiplicative:

$$f(am) = af(m)$$

for all $a \in A, m \in M$.

Note:
I'm going to skip the rest of the standard definitions, for Chrissake.

1.4.3 Definition

If M is an A-module, we define $\operatorname{ann}_A(M) \leq A$ the **annihilator** of M by

$$\operatorname{ann}_A(M) = \{a \in A \mid aM = 0\} = \ker\varphi,$$

where φ is the module map $\varphi\colon A \to \operatorname{end}(M)$. This is then an ideal. For $S \subseteq M$ a subset we define $\operatorname{ann}_A(S)$ similarly.
A module M is **faithful** if $\operatorname{ann}_A(M) = 0$.

$\operatorname{ann}_A(S)$ is equal to $\operatorname{ann}_A(\langle S\rangle)$, the annihilator of the submodule generated by S (all R-combinations of elements from S), and is thus also an ideal.

1.4.4 Definition

For a ring A, an **A-algebra** is a ring B equipped with a ring morphism $\varphi\colon A \to B$.

Note that B can be embedded into $\operatorname{end}(B)$ mapping $b \in B$ to multiplication by b. Thus an A-algebra is an A-module with multiplication given by $a \cdot b = \varphi(a)b$.

1.4.5 Definition

For two A-algebras B, C, an **A-algebra morphism** is a map $f\colon B \to C$ which is both a ring morphism and an A-module morphism.

If $\varphi\colon A \to B, \psi\colon A \to C$ are the algebra maps, then $f\colon B \to C$ is an algebra morphism iff it is a ring morphism and $\psi = f \circ \varphi$:

$$\psi(a) = a1_C = af(1_B) = f(a1_B) = f \circ \varphi(a).$$

1.4.6 Definition

For an A-module M and a submodule N, its quotient module M/N is the set of cosets $m+N$ with the usual M-module structure. The usual isomorphism results hold.

1.4.7 Proposition

Let $I \leq A$ be a module, then every A/I-module is an A-module with $\operatorname{ann}_A(M) \supseteq I$ and vice-versa. In particular an A-module M can be viewed as a faithful $A/\operatorname{ann}_A(M)$-module.

Proof

Consider the projection $\pi\colon A \to A/I$. Composing with $\varphi\colon A/I \to \operatorname{end}(M)$ gives an A-module structure to an A/I-module and $\ker(\varphi \circ \pi) \supseteq \ker \varphi = I$. Conversely if $\varphi\colon A \to \operatorname{end}(M)$ has kernel containing I we can quotient it by I. ■

1.4.8 Definition

For a collection of A-modules $\{M_\alpha\}_\alpha$ we define their **direct product** and **direct sum** by

$$\prod_\alpha M_\alpha = \{(m_\alpha) \mid m_\alpha \in M_\alpha\}, \qquad \bigoplus_\alpha M_\alpha = \left\{ (m_\alpha) \in \prod_\alpha M_\alpha \;\middle|\; \text{All but finitely many of } m_\alpha \text{ are zero} \right\}.$$

$\prod_\alpha M_\alpha$'s module structure is given by pointwise operations, and $\bigoplus_\alpha M_\alpha$ is viewed as a submodule. These form the products and coproducts in the category of A-modules.

1.4.9 Definition

An A-module M is **free** if

$$M \cong A^{\oplus S} = \bigoplus_{\alpha \in S} A$$

for some set S. Note that $\{e_\alpha\}_{\alpha \in S}$ (the usual standard basis vectors) forms a basis of $A^{\oplus S}$ in the sense that every element of $A^{\oplus S}$ can be written as a unique combination of standard basis vectors. A collection of elements $\{m_\alpha\}_{\alpha \in S}$ is a **basis** of M if mapping $m_\alpha \mapsto e_\alpha$ extends to an isomorphism $M \cong A^{\oplus S}$.

A basis of M is equivalently a set of elements which generate M and are also linearly independent in the usual way.

A free module is called such because they are free in the usual sense with respect to the obvious forgetful functor $\mathrm{Mod}_A \to \mathrm{Set}$. That is, $\hom_A(A^{\oplus S}, B) \cong \hom_{\mathrm{Set}}(S, B)$.

1.4.10 Proposition

If $A^{\oplus S} \cong A^{\oplus S'}$ then S and S' have the same cardinality.

Proof

Let $\mathfrak{m} \leq A$ be maximal, then the isomorphism $A^{\oplus S} \cong A^{\oplus S'}$ induces an isomorphism $(A/\mathfrak{m})^{\oplus S} \cong (A/\mathfrak{m})^{\oplus S'}$.

$A/\mathfrak{m}$ is a field, and so this is an isomorphism of $A/\mathfrak{m}$-modules (vector spaces) of dimensions $|S|, |S'|$. Since the dimension is invariant, their cardinalities must be equal. ■

1.4.11 Definition

If M is a free A-module, then its **rank** is the cardinality of S where $M \cong A^{\oplus S}$.

1.4.12 Definition

An A-module M is **finitely generated** (**fg**) if there is a finite set $S \subseteq M$ such that $M = \langle S \rangle$.

1.4.13 Proposition

An A-module is finitely generated iff it is a quotient of A^n for some n (up to isomorphism).

Proof

If $\varphi\colon A^n \to M$ is surjective, then $\varphi(e_i)$ generate M clearly and so it is finitely-generated. Thus if $M \cong A^n/N$ is a quotient of A^n then the projection $\pi\colon A^n \to A^n/N \cong M$ is surjective and thus M is finitely-generated. Conversely let m_i generate M, and define $\varphi\colon A^n \to M$ by $\varphi(e_i) = m_i$ which extends by linearity to all of A^n since it is free. This is surjective and then $A^n/\ker\varphi \cong M$ as desired. ■

1.4.14 Proposition

Let M be a fg A-module, and $I \leq A$ an ideal such that $M = IM$. Then there is an $x \in I$ such that $(1-x)M = 0$ ($xm = m$ for all $m \in M$).

Proof

Let $m_1, \dots, m_n$ generate M, we induct on n. For $n = 0$ we have $M = 0$ so this is clear. Consider $N = M/(m_n)$, then $IN = (IM + (m_n))/(m_n) = M/(m_n) = N$ and N is generated by $n-1$ elements, so by induction there is an $x \in I$ such that $(1-x)N = 0$. Meaning $(1-x)M \subseteq (m_n)$, and since $m_n \in M = IM$ we have

$$(1-x)m_n \in (1-x)IM = I(1-x)M \subseteq I(m_n),$$

so $(1-x)m_n = ym_n$ for some $y \in I$ and thus $(1-x-y)m_n = 0$. Now $(1-x-y)(1-x) \in 1+I$ and satisfies

$$(1-x-y)(1-x)M \subseteq (1-x-y)(m_n) = 0$$

as desired. ■

1.4.15 Corollary (Nakayama's lemma)

Let M be a fg A-module, such that $J(A)M = M$, then $M = 0$.

Proof

By above there is an $x \in J(A)$ such that $(1-x)M = 0$. Since $x \in J(A)$, $1-x \in A^\times$ so that $(1-x)M = M = 0$ as desired. ■

1.4.16 Corollary

Let M be a fg A-module and $N \subseteq M$ a submodule such that $M = IM + N$ for some $I \subseteq J(A)$, then $M = N$.

Proof

Consider M/N, then $I(M/N) = (IM+N)/N = M/N$, so by above $M/N = 0$ meaning $M = N$. ■

1.4.17 Corollary

Let $(A, \mathfrak{m})$ be local, M a fg A-module. Then $x_1, \dots, x_n \in M$ generate M iff $\overline{x}_1, \dots, \overline{x}_n \in M/\mathfrak{m}M$ generate $M/\mathfrak{m}M$ over $A/\mathfrak{m}$.

Proof

If x_i generate M then $\overline{x}_i$ clearly generate M. Conversely, let $N = (x_1, \dots, x_n)$ be the submodule generated by M, and consider the composition $f\colon N \to M \to M/\mathfrak{m}M$. f's image is the span of $(\overline{x}_i)_i$, so all of $M/\mathfrak{m}$. Thus f is surjective, so $M = N + \mathfrak{m}M$, since for every $m \in M$ there is an $n \in N$ such that $f(n) - m \in \mathfrak{m}M$ by surjectivity, and by above since $J(A) = \mathfrak{m}$ this means $M = N$ as desired. ■

1.4.18 Definition

Recall the definition of an exact sequence:

$$M \xrightarrow{f} N \xrightarrow{g} P$$

is **exact** at N if $\operatorname{im} f = \ker g$.

1.5 Localization of modules

1.5.1 Definition

Let $S \subseteq A$ be multiplicatively closed and M an A-module. Define an equivalence on $M \times S$ by $(m, s) \sim (m', s')$ iff there is a $t \in S$ such that $t(s'm - sm') = 0$. This is an equivalence relation and we denote the equivalence class of (m, s) by m/s. The quotient $S^{-1}M$ forms an A-module by

$$m/s + m'/s' = (s'm + sm')/(ss'), \qquad a \cdot (m/s) = (am)/s.$$

Note that localization at S forms an additive functor $S^{-1}\colon \mathrm{Mod}_A \to \mathrm{Mod}_{S^{-1}A}$. Indeed, $S^{-1}M$ is an $S^{-1}A$ module by

$$a/s \cdot m/s' = (am)/(ss').$$

And if $f\colon M \to N$ is an A-module morphism, then define

$$S^{-1}f\colon S^{-1}M \to S^{-1}N, \qquad S^{-1}f(m/s) = f(m)/s.$$

It is not hard to see that this is functorial and additive.

There is the canonical morphism $\iota\colon A \to S^{-1}A$ and as such it lifts to a "forgetful" functor $\mathrm{Mod}_{S^{-1}A} \to \mathrm{Mod}_A$, where for $M \in \mathrm{Mod}_{S^{-1}A}$ we view it as an A-module by $a \cdot M = \iota(a) \cdot M$.

1.5.2 Proposition

The functor $S^{-1}\colon \mathrm{Mod}_A \to \mathrm{Mod}_{S^{-1}A}$ is left adjoint to the forgetful functor $\mathrm{Mod}_{S^{-1}A} \to \mathrm{Mod}_A$.

Proof

We need to show that there is a canonical bijection

$$\hom_{S^{-1}A}(S^{-1}M, N) \cong \hom_A(M, N),$$

for all A-modules M and $S^{-1}A$-modules N. For $f\colon M \to N$ map it to $\hat{f}\colon S^{-1}M \to N$ by $\hat{f}(m/s) = 1/s \cdot f(m)$. And for $g\colon S^{-1}M \to N$ map it to $\tilde{g}\colon M \to N$ by $\tilde{g}(m) = g(m/1)$. ∎

1.5.3 Definition

An additive functor $F\colon \mathrm{Mod}_A \to \mathrm{Mod}_B$ is

(1) **left exact** if it preserves left-exact sequences: if

$$0 \to M \xrightarrow{f} N \xrightarrow{g} P$$

is exact, then so is

$$0 \to FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP.$$

(2) **right exact** if it preserves right-exact sequences: if

$$M \xrightarrow{f} N \xrightarrow{g} P \to 0$$

is exact, then so is

$$FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP \to 0.$$

(3) **exact** if it is both left and right exact.

1.5.4 Proposition

An additive functor is

(1) left exact iff it preserves kernels iff it preserves finite limits iff it preserves the left side of a short exact sequence.

(2) right exact iff it preserves cokernels iff it preserves finite colimits iff it preserves the right side of a short exact sequence.

(3) exact iff it preserves finite limits and colimits iff it preserves short exact sequences iff it preserves all exact sequences.

Proof

Left-exact sequences are kernel diagrams, and right-exact sequences are cokernel diagrams. So by definition, left exact functors are those which preserve kernels, and right exact functors are those which preserve cokernels. Since additive functors preserve finite biproducts, left/right exact functors preserve limits/colimits.

Clearly left-exact functors preserve the left side of a short exact sequence (similarly for right-exact functors). To show the converse, if left-exact sequences can be quotiented to give an appropriate short exact sequence, similar for right-exact sequences. ■

1.5.5 Corollary

Left adjoints are right exact, and right adjoints are left exact.

Proof

Left adjoints preserve all colimits, and right adjoints all limits. ■

1.5.6 Corollary

S^{-1} is an exact functor.

Proof

By the above corollary, it is right-exact. To show that it is exact, it is sufficient to show that it preserves injections. If $f\colon M \to N$ is injective then $m/s \in \ker S^{-1}f$ iff $f(m)/s = 0/1$, so that $tf(m) = 0$ for some $t \in S$. But then $tf(m) = f(tm)$ so $tm \in \ker \varphi$, so $tm = 0$, which means $m/s = 0/1$ as desired. ■

In general if $U\colon \mathrm{Mod}_A \to \mathrm{Mod}_B$ is a restriction functor corresponding to a morphism $\iota\colon B \to A$, then it has a left adjoint via the tensor product. In particular, viewing A as a (A, B)-module, then $A \otimes_B M$ has an A-module structure (given by $a(a' \otimes m) = (aa') \otimes m$). U's left adjoint is then $A \otimes_B M$.

Since left adjoints are unique up to natural isomorphism,

$$S^{-1}M \cong S^{-1}A \otimes_A M.$$

Thus $S^{-1}A \otimes_A \bullet$ is an exact functor, which means that $S^{-1}A$ is a **flat** A-module. This can also be shown directly: $S^{-1}A$ is the filtered colimit of free A-modules, and is thus flat.

1.5.7 Corollary

S^{-1} preserves arbitrary coproducts:

$$S^{-1} \bigoplus_\alpha M_\alpha \cong \bigoplus_\alpha S^{-1} M_\alpha$$

naturally.

Proof

Left adjoints preserve arbitrary colimits. ■

Note:
While S^{-1} is exact, it is not **continuous**: it does not preserve arbitrary limits (equivalently it does not preserve arbitrary products). That is, we do *not* have that

$$S^{-1}\prod_\alpha M_\alpha \cong \prod_\alpha S^{-1}M_\alpha.$$

This is of course true when the index set is finite, since then it is just a biproduct.

1.5.8 Corollary

(1) If $N \subseteq M$ is a submodule, $S^{-1}N$ can be naturally embedded in $S^{-1}M$.

(2) There is a natural isomorphism $S^{-1}(M/N) \cong (S^{-1}M)/(S^{-1}N)$.

(3) For $f\colon M \to N$ of A-modules, we have natural isomorphisms

$$\ker S^{-1}f \cong S^{-1}\ker f, \qquad \mathrm{im} S^{-1}f \cong S^{-1}\mathrm{im} f, \qquad S^{-1}\mathrm{coker} f \cong \mathrm{coker} S^{-1}f.$$

(4) For every two submodules $M_1, M_2 \subseteq M$ under the natural embeddings,

$$S^{-1}(M_1 \cap M_2) = S^{-1}M_1 \cap S^{-1}M_2 \subseteq S^{-1}M.$$

Proof

The first point is due to S^{-1} being exact and thus preserving injections. The second point is because S^{-1} preserves the short exact sequence $0 \to N \to M \to M/N \to 0$. The third point is clear as well, with only im not being immediately obvious. $\mathrm{im} f$ can be characterized as $\mathrm{im} f = \ker \mathrm{coker} f$, so it follows.

For the last point, note that $M_1 \cap M_2$ is the kernel of $M_1 \oplus M_2 \to M$ by $(x, y) \mapsto x - y$ (via the diagonal embedding), and so this follows from S^{-1}'s exactness. ■

1.5.9 Definition
For a prime ideal $\mathfrak{p} \leq A$ and an A-module M, denote by $M_\mathfrak{p}$ the localization of M by $S = A - \mathfrak{p}$.

1.5.10 Lemma
For an A-module M, consider the natural map

$$\varphi\colon M \to \prod_\mathfrak{m} M_\mathfrak{m}, \qquad m \mapsto (m/1)_\mathfrak{m},$$

where the product is over all maximal ideals of A. Then φ is injective.

Proof

Assume there is an $0 \neq x \in M$ such that $\varphi(x) = 0$. Then define

$$I = \operatorname{ann}_A(x) = \{a \in A \mid ax = 0\}.$$

I is a proper ideal of A, and thus is contained in a maximal ideal $\mathfrak{m}$. Then for every $s \in A - \mathfrak{m}$ we have that $s \notin I$ and so $sx \neq 0$, and thus $m/1$ is not zero in $M_{\mathfrak{m}}$, a contradiction. ∎

1.5.11 Corollary

For an A-module M, the following are equivalent:

(1) $M = 0$,

(2) $S^{-1}M = 0$ for every multiplicative set $S \subseteq A$,

(3) $M_{\mathfrak{p}} = 0$ for every prime ideal $\mathfrak{p} \leq A$,

(4) $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m} \leq A$.

Proof

Clearly each point implies the next. By above M embeds into $\prod_{\mathfrak{m}} M_{\mathfrak{m}}$, so if $M_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$ then $M = 0$; the last point implies the first. ∎

1.5.12 Corollary

For a morphism of A-modules $f\colon M \to N$, the following are equivalent:

(1) f is injective/surjective/an isomorphism,

(2) $S^{-1}f$ is injective/surjective/an isomorphism for every multiplicative $S \subseteq A$,

(3) $f_{\mathfrak{p}}$ is injective/surjective/an isomorphism for every prime $\mathfrak{p} \leq A$,

(4) $f_{\mathfrak{m}}$ is injective/surjective/an isomorphism for every maximal $\mathfrak{m} \leq A$.

Proof

By S^{-1}'s exactness, the first point implies the second, and clearly all the other points imply the next. To show the last implies the first, let $K = \ker f$ so that $0 \to K \to M \xrightarrow{f} N$ is exact. Then $0 \to K_{\mathfrak{m}} \to M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} N_{\mathfrak{m}}$ is exact, and so $K_{\mathfrak{m}} = \ker f_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$. By above this means $K = 0$ so that f is injective. Similar for surjectivity. ∎

1.6 Finitely generated, integral, and finite algebras

1.6.1 Definition

For an A-algebra B, and $S \subseteq B$ denote by $A[S]$ the smallest A-subalgebra containing S in B.

Let $\varphi\colon A \to B$ be the algebra map, then $A[S]$ is the subring of B generated by $\varphi(A) \cup S$. Alternatively, for every $s \in S$ define an indeterminant x_s, and let $X = \{x_s\}_{s\in S}$ be the set of all these indeterminants. Consider the polynomial algebra $A[X]$, and define $f\colon A[X] \to B$ by $f(x_s) = s$. Then $A[S]$ is just the image of f. Explicitly, for a function $\alpha\colon S \to \mathbb{Z}_{\geq 0}$ with finite support, define $S^\alpha = \prod_{\alpha(s)\neq 0} s^{\alpha(s)}$. Then

$$A[S] = \left\{ \text{Finite sums } \sum_\alpha \varphi(a_\alpha) S^\alpha \;\middle|\; a_\alpha \in A \right\}.$$

Note that we can consider B as an A-module. Then for $S \subseteq B$, we can consider the *submodule* generated by S: $\langle S\rangle$. Since $A[S]$ is itself a module, $\langle S\rangle \subseteq A[S]$ automatically, and we do not have equality in general.

1.6.2 Definition

An A-algebra B is **finitely generated** if there is a finite $S \subseteq B$ such that $B = A[S]$. B is a **finite algebra** if it is finitely generated as an A-module.

A finite A-algebra is also finitely generated as an A-algebra, but not vice versa in general.

1.6.3 Example

Suppose $S = \{s\}$ is a singleton, then $A[s]$ is the set of polynomials $\sum_k \varphi(a_k)s^k$.

1.6.4 Example

Clearly the polynomial algebra $A[x_1, \ldots, x_n]$ is a finitely generated A-algebra. But it is not a finite A-algebra: we can reduce this to the case of fields, in which x_i^k are all linearly independent and thus there is no finite basis.

1.6.5 Lemma

An A-algebra B is fg iff it is the quotient of some finite polynomial algebra $A[x_1, \ldots, x_n]$ (up to isomorphism of A-algebras).

Proof

This is equivalent to saying B is fg iff there is a surjection $A[x_1, \ldots, x_n] \to B$. This is clearly true: if B is fg then for $s_1, \ldots, s_n$ generating B map $x_i \mapsto s_i$ and extend by freeness. Conversely, quotients of fg algebras are fg. ∎

1.6.6 Definition

Let B be an A-algebra. An element $b \in B$ is **integral** over A if b is the root of a monic polynomial over A. That is, there exist $a_1, \ldots, a_n \in A$ such that $b^n + a_1 b^{n_1} + \cdots + a_n \cdot 1 = 0$.

$b \in B$ is **algebraic** over A if b is the root of any polynomial over A.

Clearly integral elements are algebraic. For fields the converse holds as well, as we can divide a polynomial by its leading coefficient.

1.6.7 Example

The set of integral elements in the $\mathbb{Z}$-algebra $\mathbb{Q}$ is $\mathbb{Z}$. Clearly an integer is integral, as $n \in \mathbb{Z}$ is the root of the monic $x - n$. Conversely if $b = p/q \in \mathbb{Q}$ is integral with p, q coprime then

$$b^n + a_1 b^{n-1} + \cdots + a_n 1 = 0, \qquad a_i \in \mathbb{Z}.$$

Then we have that

$$p^n + a_1 p^{n-1} q + \cdots + a_n q^n = 0$$

and so q must divide p^n, but since p, q are coprime this can only happen if $q = 1$ and $b \in \mathbb{Z}$.

1.6.8 Lemma

If M is a fg A-module, then for every $\varphi \in \mathrm{end}_A(M)$ there exist $a_1, \dots, a_n \in A$ such that

$$\varphi^n + a_1 \varphi^{n-1} + \cdots + a_n = 0$$

in $\mathrm{end}_A(M)$.

Proof

The proof is by a generalization of Cayley-Hamilton. Let $n \geq 1$ and consider $R = \mathbb{Z}[X_{ij}]$, the polynomial ring over n^2 indeterminants X_{ij}, $1 \leq i, j \leq n$. Let $X = (X_{ij}) \in M_n(R)$, and define its **characteristic polynomial**: $\Phi_X(T) \in R[T]$ given by

$$\Phi_X(T) = \det(T\mathrm{id} - X).$$

By Cayley-Hamilton over R's field of fractions, $\Phi_X(X) = 0$.

Now if A is any ring and $Y \in M_n(A)$, there is a unique ring morphism $\varphi\colon R \to A$ which maps X to Y (coordinate-wise). Indeed, such a map must map X_{ij} to Y_{ij} which uniquely determines φ. Then φ lifts to the ring morphism $R[T] \to A[T]$ (map $\sum_n a_n T^n$ to $\sum_n \varphi(a_n) T^n$), which maps $\Phi_X(T)$ to $\Phi_Y(T)$ (which is also given by the determinant $\det(T\mathrm{id} - Y)$ since the determinant is functorial: $\det(\varphi(B)) = \varphi(\det(B))$). Thus $\Phi_X(X) = 0$ implies that $\Phi_Y(Y) = \varphi(\Phi_X(\varphi(X))) = \varphi(\Phi_X(X)) = 0$, as desired. So Cayley-Hamilton applies to any commutative ring.

Now for our lemma, let $m_1, \dots, m_n \in M$ be generators. Then for all j we have

$$\varphi(m_j) = \sum_i a_{ij} m_i, \qquad a_{ij} \in A.$$

We associate to φ the representative matrix $[\varphi] = (a_{ij}) \in M_n(A)$ (which is not unique, unlike over fields). This has the usual properties that $[\varphi^n] = [\varphi]^n$ and $[\varphi + \psi] = [\varphi] + [\psi]$ (meaning that the composition or sum of representatives is a representative matrix). Now Cayley-Hamilton gives us that

$$\sum_n b_n [\varphi]^n = 0$$

for some $b_n \in A$, and thus

$$\left[\sum_n b_n \varphi^n\right] = 0,$$

and since only the zero endomorphism has the zero representative, this means $\sum_n b_n \varphi^n = 0$ as desired. ■

1.6.9 Proposition

For an A-algebra B and $b \in B$, the following are equivalent:

(1) b is integral over A,

(2) $A[b] \subseteq B$ is a finite algebra (fg module) over A,

(3) $A[b]$ is contained in a finite algebra over A,

(4) There exists a faithful $A[b]$-module M such that M is finitely generated as an A-module.

Proof

Clearly (2) implies (3) implies (4): take $M = C$, and since $A[b] \subseteq M$ it is faithful (the algebra map is inclusion, which is injective). (1) implies (2) since if $b^n + a_1 b^{n-1} + \cdots + a_n 1 = 0$ then b^n is in $\langle 1, \ldots, b^{n-1} \rangle \subseteq B$ and then by induction so is every higher power of b, so that $A[b] = \langle 1, \ldots, b^{n-1} \rangle$ is finite.

(4) implies (1): we have a ring morphism $\varphi\colon A[b] \to \mathrm{end}_A(M)$, so that every $f \in A[b]$ gives us an endomorphism $\varphi(f) \in \mathrm{end}_A(M)$. In particular by the above lemma there are $a_1, \ldots, a_n \in A$ such that

$$\rho(b)^n + a_1 \rho(b)^{n-1} + \cdots + a_n = 0 \in \mathrm{end}_A(M).$$

For $f(b) = b^n + a_1 b^{n-1} + \cdots + a_n \in A[b]$, we see that $\rho(f(b)) = 0$. But M is a faithful $A[b]$-module, so $f(b) = 0$ and b is integral as desired. ∎

1.6.10 Lemma

(1) If B is a finite A-algebra, and M is a fg B-module, then M is a fg A-module.

(2) If B is a finite A-algebra, and C is a finite B-algebra, then C is a finite A-algebra.

Proof

Clearly the second point follows from the first. Now if M is generated by $m_1, \ldots, m_n$ over B, and B is generated by $b_1, \ldots, b_k$ over A then $b_i m_j$ generates M over A. Indeed, for $m \in M$ if $m = \sum_i b'_i m_i$, then $b'_i = \sum_j a_{ij} b_j$ so

$$m = \sum_{i,j} a_{ij}(b_j m_i),$$

as desired. ∎

1.6.11 Corollary

Let B be an A-algebra, and suppose that $b_1, \ldots, b_n \in B$ are integral over A. Then $A[b_1, \ldots, b_n]$ is a finite A-algebra.

Proof

By induction: for $n = 1$ this is our above proposition. Then if b_{n+1} is integral over A, it is integral over $A[b_1, \ldots, b_n]$ so that $A[b_1, \ldots, b_{n+1}]$ is a finite $A[b_1, \ldots, b_n]$-algebra, and thus a finite A-algebra by transitivity which we just showed. ∎

1.6.12 Corollary

Let B be an A-algebra, then

$$B_0 = \{b \in B \mid b \text{ is integral over } A\}$$

is an A-subalgebra of B

Proof

Clearly $0, 1 \in B_0$. Now if $b_1, b_2 \in B_0$ then $A[b_1 + b_2], A[b_1 b_2] \subseteq A[b_1, b_2]$ which is a finite A-algebra and thus $b_1 + b_2, b_1 b_2$ are integral over A; B_0 is a subring of B. And if $a \in A, b \in B_0$ then $A[ab] \subseteq A[b]$ is finite, so that $ab \in B_0$. ■

1.6.13 Definition

(1) For an A-algebra B, we define A's **integral closure** in B to be the set of all integral elements over A in B; which we showed above is an A-subalgebra.

(2) An A-algebra B is **integral** if every element of B is integral over A.

(3) A morphism of rings $f\colon A \to B$ is **integral** if B is an integral A-algebra via f.

(4) For a morphism of rings $f\colon A \to B$, $f(A)$ is **integrally closed** in B if $B_0 = f(A)$ (i.e. no $b \in B - f(A)$ is integral over A).

(5) A domain A is **normal** if it is integrally closed in its field of fractions.

1.6.14 Example

We saw that $\mathbb{Z}$ is normal (its field of fractions is $\mathbb{Q}$); the same proof shows that any UFD is normal. In particular $k[x_1, \ldots, x_n]$ and $\mathbb{Z}[x_1, \ldots, x_n]$ are normal.

Consider the k-subalgebra $A = k[x^2, x^3] \subseteq k[x]$. Then $x \notin A$ is integral over A since it is a root of $t^6 - x^6 \in A[t]$, and $x = x^3/x^2$ is in its field of fractions. Thus A is not normal.

Note:

There is similarity between integral morphisms and algebraic extensions of fields. But while algebraically closed fields exist, there are no "integrally closed ring": every ring can be embedded non-integrally into another ring.

1.6.15 Corollary

An A-alegbra B is finite over A iff it is integral and finitely generated.

Proof

If B if a finite A-algebra, then for every $b \in B$, $A[b] \subseteq B$ is finite so b is integral over A; B is integral and finitely generated. Conversely if B is integral and finitely generated, suppose $B = A[b_1, \ldots, b_n]$; since each b_i is integral we have that B is finite. ■

1.6.16 Corollary

Let B be an integral A-algebra, and C be a B-algebra.

(1) If $c \in C$ is integral over B, then it is integral over A.

(2) If C is integral over B, then it is integral over A.

Proof

If c is integral over B, then

$$c^n + b_1 c^{n-1} + \cdots + b_n = 0$$

for some $b_i \in B$, which are integral over A. Then $B' = A[b_1, \ldots, b_n] \subseteq B$ is a finite A-algebra, and c is integral over B' and so $B'[c]$ is a finite B'-algebra. Since B' is a finite A-algebra, by transitivity $B'[c]$ is a finite A-algebra as well containing $A[c]$, so c is integral over A.

The second point follows from the first. ■

1.6.17 Corollary

Let B be an A-algebra, and B_0 be the integral closure of A inside B. Then B_0 is integrally closed in B.

Proof

If $b \in B$ is integral over B_0, then by the above corollary, since B_0 is an integral A-algebra, b is integral over A and thus $b \in B_0$. ■

Let B be an A-algebra under $f\colon A \to B$.

(1) For an ideal $J \leq B$, define $J_A = f^{-1}(J) \leq A$. Then f induces an injective homomorphism $A/J_A \to B/J$, so that B/J is an A/J_A-algebra.

(2) For a multiplicative set $S \subseteq A$, $f(S) \subseteq B$ is multiplicative. Then we have a natural morphism $S^{-1}A \to f(S)^{-1}B$ mapping a/s to $f(a)/f(s)$. Thus $f(S)^{-1}B$ is an $S^{-1}A$-algebra.

(3) Viewing B as an A-module, we have already defined $S^{-1}B$, and this clearly coincides with $f(S)^{-1}B$. Thus $S^{-1}B$ is naturally an $S^{-1}A$-algebra.

1.6.18 Proposition

Let B be an integral A-algebra.

(1) Let $J \leq B$ be an ideal, then B/J is an integral A/J_A-algebra.

(2) Let $S \subseteq B$ be multiplicative, then $S^{-1}B$ is an integral $S^{-1}A$-algebra.

Proof

For $\overline{x} \in B/J$, lift it to $x \in B$, which is integral over A. Thus $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ for $a_i \in A$, and then

$$\overline{x}^n + \overline{a}_1 \overline{x}^{n-1} + \cdots + \overline{a}_n = 0,$$

where $\overline{a}_i = a_i + J_A \in A/J_A$, so that $\overline{x}$ is integral over A/J_A.

An element of $S^{-1}B$ is of the form $\tilde{x} = x/f(s)$ with $x \in B, s \in S$. By assumption, $x^n + a_1x^{n-1} + \cdots + a_n = 0$ and thus

$$\left(\frac{x}{f(s)}\right)^n + \frac{a_1}{s}\left(\frac{x}{f(s)}\right)^{n-1} + \cdots + \frac{a_n}{s^n} = \frac{x^n + a_1x^{n-1} + \cdots + a_0}{f(s)^n} = 0,$$

so that $\tilde{x}$ is integral over $S^{-1}A$ as desired. ■

1.6.19 Proposition

Let $A \subseteq B$ be integral domains so that B is integral over A. Then A is a field iff B is a field.

Proof

If A is a field, then let $0 \neq b \in B$. It is integral over A so $b^n + a_1b^{n-1} + \cdots + a_n = 0$ for $a_i \in A$, and assume n is minimal. Then

$$b(b^{n-1} + a_1b^{n-2} + \cdots + a_{n-1}) = -a_n.$$

Since by minimality $b^{n-1} + a_1b^{n-1} + \cdots + a_{n-1} \neq 0$, we have $a_n \neq 0$. Thus $-a_n$ is a unit, and thus b must be a unit as well (if $xy = u$ is a unit, then x is a unit since $x(yu^{-1}) = 1$). So B is a field.

Conversely if B is a field, then let $0 \neq x \in A$. Then $x^{-1} \in B$ since it is a field. Since B is integral over A, there are $a_i \in A$ such that

$$x^{-n} + a_1x^{-(n-1)} + \cdots + a_n = 0,$$

and multiplying by x^{n-1} we get

$$x^{-1} + a_1 + \cdots + a_nx^{n-1},$$

so that x^{-1} is the sum of elements from A, and thus is in A. ■

1.6.20 Corollary

Let B be an integral A-algebra, and $\mathfrak{q} \leq B$ a prime ideal. Let $\mathfrak{p} = \mathfrak{q}_A \leq A$ be its preimage, then $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is maximal.

Proof

Both $\mathfrak{p}, \mathfrak{q}$ are prime so that $B/\mathfrak{q}$ and $A/\mathfrak{p}$ are integral domains. As we showed, $B/\mathfrak{q}$ is integral over $A/\mathfrak{p}$. Thus $B/\mathfrak{q}$ is a field iff $A/\mathfrak{p}$ is a field; so $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is maximal. ■

1.6.21 Corollary

Let B be an integral A-algebra, $\mathfrak{q} \leq \mathfrak{q}' \leq B$ be prime ideals such that $\mathfrak{q}_A = \mathfrak{q}'_A$. Then $\mathfrak{q} = \mathfrak{q}'$.

Proof

Let $\mathfrak{p} = \mathfrak{q}_A = \mathfrak{q}'_A$; it is prime over A. Consider the following commutative square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\iota_A"'] & B \arrow[d, "\iota_B"] \\ A_{\mathfrak{p}} \arrow[r, "f_{\mathfrak{p}}"] & B_{\mathfrak{p}} \end{tikzcd}$$

which commutes since the structure morphism $A \to S^{-1}A$ is a natural transformation (literally what the commutative square says).

Since $f^{-1}\mathfrak{q} = \mathfrak{p}$ we see that $\mathfrak{q} \cap f(A - \mathfrak{p}) = \emptyset$ and similarly for $\mathfrak{q}'$. Thus $\mathfrak{q}B_{\mathfrak{p}} \subseteq \mathfrak{q}'B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ are prime ideals (since they don't intersect the localizing set $f(A - \mathfrak{p})$).

We are going to show that $\mathfrak{q}B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ is a maximal ideal so that $\mathfrak{q}B_{\mathfrak{p}} = \mathfrak{q}'B_{\mathfrak{p}}$. And then since $\mathfrak{q} \mapsto \mathfrak{q}B_{\mathfrak{p}}$ is bijective with inverse ι_B^{-1}, we will have that $\mathfrak{q} = \mathfrak{q}'$.

Since B is integral over A, $B_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$. Thus, by the above corollary, it is sufficient to show that $\tilde{\mathfrak{p}} = f_{\mathfrak{p}}^{-1}(\mathfrak{q}B_{\mathfrak{p}}) \subseteq A_{\mathfrak{p}}$ is maximal. By assumption,

$$\iota_A^{-1}(\tilde{\mathfrak{p}}) = \iota_A^{-1}(f_{\mathfrak{p}}^{-1}(\mathfrak{q}B_{\mathfrak{p}})) = f^{-1} \circ \iota_B^{-1}(\mathfrak{q}B_{\mathfrak{p}}) = f^{-1}(\mathfrak{q}) = \mathfrak{p}.$$

Thus we have that $\tilde{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ (since ι_A^{-1} is bijective) which is maximal, as desired. ∎

The main result we want is the following.

1.6.22 Theorem

Let $f\colon A \to B$ be an injective integral morphism. Then for every prime $\mathfrak{p} \leq A$, there exists a prime ideal $\mathfrak{q} \leq B$ such that $\mathfrak{p} = \mathfrak{q}_A$.

Proof

Since $B_{\mathfrak{p}} = f(A - \mathfrak{p})^{-1}B$, and $0 \notin f(A - \mathfrak{p})$ because f is injective, $B_{\mathfrak{p}} \neq 0$. As before, $B_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$ and we have the same commutative square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\iota_A"'] & B \arrow[d, "\iota_B"] \\ A_{\mathfrak{p}} \arrow[r, "f_{\mathfrak{p}}"] & B_{\mathfrak{p}} \end{tikzcd}$$

Let $\mathfrak{m} \subseteq B_{\mathfrak{p}}$ be maximal, so that $f_{\mathfrak{p}}^{-1}(\mathfrak{m}) \subseteq A_{\mathfrak{p}}$ is maximal (as we showed $\mathfrak{q}_A$ is maximal iff $\mathfrak{q}$ is). Thus $f_{\mathfrak{p}}^{-1}(\mathfrak{m}) = \mathfrak{p}A_{\mathfrak{p}}$. Let $\mathfrak{q} = \iota_B^{-1}(\mathfrak{m})$, which is prime in B, and

$$f^{-1}(\mathfrak{q}) = f^{-1} \circ \iota_B^{-1}(\mathfrak{m}) = \iota_A^{-1} \circ f_{\mathfrak{p}}^{-1}(\mathfrak{m}) = \iota_A^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p},$$

as desired. ∎

Note:

Equivalently, this theorem says that if $f\colon A \to B$ is integral injective, then $f^*\colon \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is surjective.

Note that injectivity is necessary here. This is because $\pi\colon A \to A/I$ is integral for any ideal $I \leq A$, but $\pi^*\colon \mathrm{Spec}(A/I) \to \mathrm{Spec}(A)$ is not surjective: its image is $V(I)$, the set of prime ideals containing I.

1.6.23 Corollary

Let B be an integral A-algebra, then for every prime $\mathfrak{q} \leq B$ and every prime ideal $\mathfrak{p}' \supseteq \mathfrak{p} = \mathfrak{q}_A$, there exists a prime ideal $\mathfrak{q}' \supseteq \mathfrak{q}$ of B such that $\mathfrak{q}'_A = \mathfrak{p}'$.

Proof

Consider the commutative square

$$\begin{tikzcd} A \arrow[r, "f"'] \arrow[d, "\pi_A"'] & B \arrow[d, "\pi_B"'] \\ A/\mathfrak{p} \arrow[r, "\overline{f}"] & B/\mathfrak{q} \end{tikzcd}$$

Then $\mathfrak{p}' \supseteq \mathfrak{p}$ gives rise to a prime ideal $\overline{\mathfrak{p}}' = \mathfrak{p}'/\mathfrak{p} \subseteq A/\mathfrak{p}$. Since $B/\mathfrak{q}$ is an integral $A/\mathfrak{p}$-algebra, there is a prime ideal $\overline{\mathfrak{q}}' \subseteq B/\mathfrak{q}$ such that $\overline{f}^{-1}(\overline{\mathfrak{q}}') = \overline{\mathfrak{p}}'$. Then $\mathfrak{q}' = \pi_B^{-1}(\overline{\mathfrak{q}}')$ is a prime ideal of B containing $\mathfrak{q}$ and

$$f^{-1}(\mathfrak{q}') = f^{-1} \circ \pi_B^{-1}(\overline{\mathfrak{q}}') = \pi_A^{-1} \circ \overline{f}^{-1}(\overline{\mathfrak{q}}') = \pi_A^{-1}(\overline{\mathfrak{p}}') = \mathfrak{p}',$$

as desired. ∎

1.6.24 Proposition

Let B be an A-algebra, $C \subseteq B$ be the integral closure of A inside of B. Then for any multiplicative $S \subseteq A$, $S^{-1}C$ is the integral closure of $S^{-1}A$ inside of $S^{-1}B$.

Proof

Since C is integral over A, $S^{-1}C$ is integral over $S^{-1}C$. Now if b/s is integral over $S^{-1}A$, then $b/1 = s \cdot b/s$ is integral over $S^{-1}A$ as well, and thus

$$\left(\frac{b}{1}\right)^n + \frac{a_1}{s_1}\left(\frac{b}{1}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0$$

for $a_i/s_i \in S^{-1}A$. Let $t = s_1 \cdots s_n$ and $u_i = t/s_i \in S$, and multiplying this equation by t^n gives us

$$\left(\frac{tb}{1}\right)^n + a_1u_1\left(\frac{bt}{1}\right)^{n-1} + \cdots + a_nu_nt^{n-1} = 0.$$

Thus there is a $v \in S$ such that

$$v((bt)^n + a_1u_1(bt)^{n-1} + a_2u_2t(bt)^{n-2} + \cdots + a_nu_nt^{n-1}) = 0.$$

Then btv is integral over A (multiply by v^{n-1}), so that $btv \in C$. Then $b/s = (btv)/(stv) \in S^{-1}C$, as desired. ∎

1.6.25 Corollary

Let A be a domain, then the following are equivalent:

(1) A is normal,

(2) $S^{-1}A$ is normal for every multiplicative $S \subseteq A$ not containing 0,

(3) $A_{\mathfrak{p}}$ is normal for every prime $\mathfrak{p}$,

(4) $A_{\mathfrak{m}}$ is normal for every maximal $\mathfrak{m}$.

Proof

Let K be A's field of fractions, and let $C \subseteq K$ be A's integral closure in K. Then $S^{-1}C$ is $S^{-1}A$'s integral closure in $S^{-1}K = K$ for all multiplicative S not containing 0. If A is normal, then $C = A$ so that $S^{-1}A$ is integrally closed in K, and thus normal. So (1) implies (2), which implies (3) which implies (4).

Now if $A_{\mathfrak{m}}$ is normal, then $A_{\mathfrak{m}} \to C_{\mathfrak{m}}$ is an isomorphism for all $\mathfrak{m}$. We showed that this implies $A \to C$ is an isomorphism, so that A is normal. ■

1.7 Noetherian rings and modules

1.7.1 Lemma

Let Σ be a poset, then the following conditions are equivalent:

(1) every nonempty subset of Σ has a maximal element,

(2) every increasing chain $x_1 \leq x_2 \leq \cdots$ in Σ eventually stabilizes: at some point $x_n = x_{n+1}$ (equivalently, there are no infinite strictly increasing chains).

A poset satisfying these equivalent conditions is said to have the **ascending chain condition** (**ACC**).

Proof

The first implies the second as we can consider the subset $\{x_i\}$. The second implies the first since if $\Sigma' \subseteq \Sigma$ doesn't have a maximal element we can form a strictly increasing chain inductively. ■

1.7.2 Definition

An A-module M is **Noetherian** if its poset of A-submodules has the ACC. A ring A is **Noetherian** if it is Noetherian as a module over itself (equivalently its ideals satisfy the ACC).

1.7.3 Example

Any field is a Noetherian ring, since it only has two ideals.

$\mathbb{Q}$ is not a Noetherian $\mathbb{Z}$-module, since $\frac{1}{n!}\mathbb{Z}$ does not stabilize (the reason for the factorial is so that it forms an increasing chain).

The ring $k[x_1, x_2, \ldots]$ with infinitely many indeterminants is not Noetherian, since $(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$ forms a strictly increasing chain.

1.7.4 Proposition

An A-module is Noetherian iff every submodule is finitely generated. Thus a ring is Noetherian iff every ideal is finitely generated.

Proof

If M is Noetherian and $N \subseteq M$ is a submodule, suppose it is not finitely generated. Then inductively we will form an infinitely increasing sequence of submodules. Let $x_1 \in N$ arbitrary, then since N is not finitely generated $x_1, \ldots, x_n$ does not generate N so that there is an $x_{n+1} \in N - \langle x_1, \ldots, x_n \rangle$. Then $N_n = \langle x_1, \ldots, x_n \rangle$ is a strictly increasing infinite chain of submodules of M, a contradiction.

Conversely if every submodule of M is finitely generated, then M is Noetherian. Otherwise, let $M_1 \subsetneq M_2 \subseteq \cdots$ be an increasing chain of submodules. Let $N = \bigcup_n M_n$ be their union, which is also a submodule of M, and thus is finitely generated. Suppose $x_1, \ldots, x_n \in N$ generate N, and since M_i form a chain there is an m such that $x_1, \ldots, x_n \in M_m$ and thus $M_m = N$ so that M_i eventually stabilizes to M_m. ■

1.7.5 Corollary

Every PID is Noetherian.

1.7.6 Lemma

Let $0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$ be a short exact sequence of A-modules. Then M is Noetherian iff M', M'' are.

Equivalently, if M is an A-module and $N \subseteq M$ a submodule, M is Noetherian iff N and M/N are.

Proof

If M is Notherian, let $M'_n \subseteq M', M''_n \subseteq M''$ be increasing sequences of submodules. Then $f(M'_n)$ and $g^{-1}(M''_n)$ are submodules of M and thus stabilize: $f(M'_n) = f(M'_{n+1})$ and $g^{-1}(M'_n) = g^{-1}(M'_{n+1})$ eventually. By the injectivity of f and surjectivity of g, this means M'_n and M''_n both eventually stabilize.

Conversely if M', M'' are Noetherian, let $M_n \subseteq M$ be an increasing sequence of submodules. Then $f^{-1}(M_n) = M_n \cap M'$ and $g(M_n) \subseteq M''_n$ are increasing sequences of submodules of M', M'' respectively. Thus eventually they stabilize: $f^{-1}(M_n) = f^{-1}(M_{n+1})$ and $g(M_n) = g(M_{n+1})$ eventually. We claim that this means $M_n = M_{n+1}$.

Indeed, for $x \in M_{n+1}$ consider $g(x) \in g(M_{n+1})$ so there is a $y \in M_n$ such that $g(x) = g(y)$. Then $x - y \in \ker g = \operatorname{im} f$ so that $x - y = f(z)$ for $z \in M'$, and $x - y \in M_{n+1}$ so that $z \in f^{-1}(M_{n+1})$ and thus $z \in f^{-1}(M_n)$. This means that $x - y \in M_n$ and thus $x \in M_n$ as desired. ■

1.7.7 Corollary

(1) The finite direct sums of Noetherian modules is Noetherian.

(2) If A is a Noetherian ring and M is a fg A-module, then M is a Notherian module.

(3) If A is a Notherian ring and B a finite A-algebra, then B is a Noetherian ring. In particular, A/I is Notherian for any ideal $I \leq A$.

Proof

For the first point it is sufficient to show this for binary direct sums. If M, N are Noetherian, then $0 \to M \to M \oplus N \to N \to 0$ is a short exact sequence and thus $M \oplus N$ is Notherian by above.

Since M is fg, $M \cong A^n/N$ for some submodule $N \leq A^n$ (consider the surjection $A^n \to M$ mapping to generators). By the first point, A^n is Noetherian and thus M is as well.

B is a fg A-module, and thus a Noetherian A-module. Since ideals of B are A-submodules, the ACC on A-submodules implies ACC on B's ideals. ■

1.7.8 Lemma

Let A be a Noetherian ring, then all of its localizations are Noetherian rings as well.

Proof

Let $S \subseteq A$ be multiplicative, and consider an increasing sequence of ideals $\tilde{I}_1 \subseteq \tilde{I}_2 \subseteq \cdots$ in $S^{-1}A$. Then $I_n = \iota^{-1}(\tilde{I}_n)$ forms an increasing sequence of ideals of A, and thus stabilizes. Since $\tilde{I}_n = S^{-1}I_n$, we get that $\tilde{I}_n$ stabilizes. ■

1.7.9 Theorem (Hilbert basis theorem)

If A is Noetherian, then $A[x]$ is a Noetherian ring.

Proof

Let $I \subseteq A[x]$ be an ideal, we wish to show it is finitely generated. Let us adopt the notation that for $f \in A[x]$, $a_f \in A$ denotes the leading term of f (and $a_f = 0$ only when $f = 0$).

We claim that $J = \{a_f \mid f \in I\}$ is an ideal in A. If $f \in I$ and $b \in A$ then either $ba_f = 0$ or $ba_f = a_{bf}$; in both cases $ba_f \in J$. Now if $f, g \neq 0$ are in I then $a_f + a_g$ is either zero or the leading term of $x^{\deg g} f + x^{\deg f} g \in J$ and thus $a_f + a_g \in J$.

Since J is Noetherian, there are $f_1, \ldots, f_n \in I$ such that $J = (a_{f_1}, \ldots, a_{f_n})$. Let $d_i = \deg f_i$, and d be the maximum d_i. Consider the fg A-submodule

$$M = \sum_{i=0}^{d-1} Ax^i \subseteq A[x].$$

Then $M \cap I$ is an A-submodule of M, which is Noetherian (since M is fg and A is Noetherian so M is Noetherian, and thus its submodules are Noetherian).

Now define $I' = (f_1, \ldots, f_n) \subseteq I$. We claim that

$$I = I' + (M \cap I) \text{ as } A\text{-submodules.}$$

Let $f \in I$, then we claim there is an $f' \in I'$ such that $\deg(f - f') < d$ and thus $f - f' \in M \cap I$. If $\deg f < d$ then we are done, so assume $\deg f \geq d$. By induction, it is sufficient to show that there is an $f' \in I'$ such that $\deg(f - f') < \deg f$.

Now, $a_f \in J = (a_{f_1}, \ldots, a_{f_n})$ so that $a_f = \sum_i b_i a_{f_i}$ for $b_i \in A$. Then define

$$f' = \sum_{i=1}^{n} b_i x^{(\deg f) - d_i} f_i \in I'.$$

By assumption, $\deg f \geq d \geq d_i$ so that $\deg(x^{(\deg f) - d_i} f_i) = d_i$, and so $\deg f' \leq \deg f$. Moreover, the leading coefficient of f' is

$$a_{f'} = \sum_{i=1}^{n} b_i a_{f_i} = a_f,$$

so that $\deg(f - f') < \deg f$ as desired.
Now since $I = I' + (M \cap I)$, I' is a fg ideal and $M \cap I$ is a fg A-module (and thus fg by $A[x]$), we have that I is a fg ideal, as desired. ■

1.7.10 Corollary

(1) If A is a Noetherian ring, so is $A[x_1, \dots, x_n]$.

(2) If A is a Noetherian ring, and B is a fg A-algebra, B is Noetherian.

(3) Any fg algebra over a field is Noetherian.

(4) Any fg algebra over a PID (in particular $\mathbb{Z}$) is Noetherian.

Proof

The first point follows by induction and Hilbert's basis theorem. The second point is because if B is a fg A-algebra, it is isomorphic to $A[x_1, \dots, x_n]/I$ for some ideal $I \leq A[x_1, \dots, x_n]$, and is thus Noetherian. The last two points are due to the second (fields and PIDs are Noetherian). ■

We now utilize these results to reprove a lemma we showed above.

1.7.11 Lemma
Let A be a ring and $T \in M_n(A)$ a matrix. Then there are $a_1, \dots, a_n \in A$ such that

$$T^n + a_1 T^{n-1} + \cdots + a_m = 0.$$

Proof

Denote $T = (a_{ij})$, and consider the subalgebra $A' = \mathbb{Z}[\{a_{ij}\}] \subseteq A$. This is a fg $\mathbb{Z}$-algebra and thus Noetherian. Since $T \in M_n(A')$, we can assume from the getgo that A is Noetherian.
Since $M_n(A) \cong A^{n^2}$ as A-modules, $M_n(A)$ is a Noetherian A-module. Thus the increasing sequence of A-modules

$$N_m = \langle 1, T, \dots, T^m \rangle \subseteq M_n(A)$$

stabilizes, so eventually $N_n = N_{n-1}$ so that $T^n \in N_{n-1}$ and we get the desired result. ■

1.7.12 Lemma
If M is a fg A-module, then for $\varphi \in \mathrm{end}_A(M)$, there are $a_1, \dots, a_n \in A$ such that

$$\varphi^n + a_1 \varphi^{n-1} + \cdots + a_n = 0.$$

Proof

When $M = A^n$ there is a natural isomorphism of A-algebras $M_n(A) \cong \mathrm{end}_A(M)$, so the result follow from the above lemma. Otherwise, consider a surjective map $\pi\colon A^n \to M$, then we claim there is a $\tilde{\varphi}\colon A^n \to A^n$

such that the below diagram commutes

$$\begin{tikzcd} A^n \arrow[r, "\tilde{\varphi}"'] \arrow[d, "\pi"'] & A^n \arrow[d, "\pi"'] \\ M \arrow[r, "\varphi"] & M \end{tikzcd}$$

Meaning $\pi \circ \tilde{\varphi} = \varphi \circ \pi$. For every i choose a preimage $V_i \in \pi^{-1}(\varphi(\pi(e_i))) \in A^n$ and define

$$\tilde{\varphi}(x_1, \dots, x_n) = \sum_{i=1}^{n} x_i v_i.$$

This satisfies the condition.
Now there is a monic polynomial $f \in A[x]$ such that $f(\tilde{\varphi}) = 0$ by the $M = A^n$ case. Then

$$f(\varphi) \circ \pi = \pi \circ f(\tilde{\varphi}) = 0,$$

which implies $f(\varphi) = 0$ because π is surjective. ∎

1.8 Simple modules and length

1.8.1 Definition

A poset Σ has the **descending chain condition** (**DCC**) if it satisfies the dual of the ACC (equivalently, Σ^{op} has ACC). This means that every subset has a minimal element, or every descending chain stabilizes.

1.8.2 Definition

An A-module is **Artinian** if it has DCC on its submodules. A ring A is **Artinian** if it is Artinian as a module over itself, equivalently it has DCC on its ideals.

Note that the short exact sequence result on Noetherian rings holds for Artinian rings as well. Thus finite direct sums of Artinian modules are Artinian, fg modules over Artinian rings are Artinian, and thus finite algebras over Artinian rings are Artinian.
Note that it is not the case that a module is Artinian iff its submodules are all fg. That is, Artinian is not equivalent to Noetherian (and for modules they are distinct conditions, for rings we will see that Artinian implies Noetherian).

1.8.3 Example

$\mathbb{Z}$ is a Noetherian but not an Artinian ring. It is not Artinian because $\mathbb{Z} \supsetneq (n) \supsetneq (n^2) \supsetneq (n^3) \supsetneq \cdots$ is an infinite decreasing sequence of ideals.
Similarly $k[x]$ is Noetherian but not Artinian: (x^k) forms a descending sequence of ideals.

1.8.4 Definition

A nonzero module M is **simple** if it has no non-trivial submodules.

1.8.5 Lemma

An A-module M is simple iff it is isomorphic (as A-modules) to $A/\mathfrak{m}$ for a maximal ideal $\mathfrak{m} \leq A$.

Proof

If M is simple, then for $0 \neq m \in M$ the submodule $\langle m \rangle \subseteq M$ is isomorphic to $A/\mathrm{ann}(m)$ (the map $a \mapsto am$ has kernel $\mathrm{ann}(m)$). Since M is simple, $\langle m \rangle = M$ and so $A/\mathrm{ann}(m) \cong M$. There is then a bijection between submodules of M and ideals of A containing $\mathrm{ann}(m)$. Since M is simple, there cannot be any nontrivial such ideals, so $\mathrm{ann}(m)$ is maximal.

And if $M \cong A/\mathfrak{m}$ then again there is a bijection between submodules of M and ideals containing $\mathfrak{m}$. So M is simple, as $\mathfrak{m}$ is maximal. ■

1.8.6 Definition

A **composition series** of M is a chain of submodules

$$0 = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots \subsetneq M_n = M$$

which is maximal. This is equivalent to M_{i+1}/M_i being simple for all i (no modules between M_i and M_{i+1}). The **length** of such a series is n (not $n+1$).

Denote by $\ell_A(M)$ the minimal length of a composition series of M (or infinity if no composition series exists).

1.8.7 Lemma

If M has finite length $\ell(M) = n$, then all of its composition series have the same length n, and every chain of submodules can be extended to a composition series.

Proof

First we claim that if $N \subsetneq M$ is a submodule, then $\ell(N) < \ell(M)$. Indeed, if $\{M_i\}$ is a composition series for N, then $N_i = M_i \cap N$ is a sequence of submodules of N. Then $N_i = N_{i+1} \cap M_i$ so

$$N_{i+1}/N_i = N_{i+1}/(N_{i+1} \cap M_i) \cong (N_{i+1} + M_i)/M_i \subseteq M_{i+1}/M_i$$

which is either simple or zero. Thus removing repeated values, we have formed a composition series for N of shorter length, so $\ell(N) \leq \ell(M)$.

Now if $\ell(N) = \ell(M)$ then $N_{i+1} + M_i = M_{i+1}$ for all i, and thus we claim $N_i = M_i$ for all i so that $N = M$. This is by induction, since $M_0 = N_0 = 0$, and $N_{i+1} = N_{i+1} + N_i = N_{i+1} + M_i = M_{i+1}$, as desired.

Now if $\{M_i\}_0^n$ is any chain in M, then

$$\ell(M) = \ell(M_n) > \ell(M_{n-1}) > \cdots > \ell(M_0) = 0,$$

so $\ell(M) \geq n$.

Now if $\{M_i\}$ is any composition series of M of length n, then $\ell(M) \geq n$ by what we just showed. But by definition $\ell(M) \leq n$ so that $\ell(M) = n$ and thus all composition series have length $\ell(M)$.

And for any general chain of submodules of M, if it is not a composition series then new terms can be inserted. This process can be repeated, and must terminate (since once it reaches a length of $\ell(M)$ it must be a composition series). ■

Note:

Note that we can prove something stronger, the **Jordan-Holder theorem**: for any two composition series M_i, N_j there is a permutation σ between i and j such that $M_{i+1}/M_i \cong N_{\sigma(i)+1}/N_{\sigma(i)}$. A proof of this can be found in my representation theory homework somewhere.

1.8.8 Lemma

An A-module M has finite length iff it is both Noetherian and Artinian.

Proof

If M has finite length, then any increasing/decreasing sequence has length bound by $\ell(M)$ so that M is both Noetherian and Artinian.

Conversley if M is Artinian and Noetherian, then consider the set of submodules $N \subseteq M$ of finite length. It is nonempty since it contains 0, and because M is Noetherian contains a maximal element M'. If $M = M'$ then we are done. Otherwise, the set of submodules $M' \subsetneq N \subseteq M$ is nonempty (containing M) and has a minimal element N since M is Artinian. But then N/M' is simple, and so composing a composition series of M' with $\subsetneq N$ forms a composition series of N, so N has finite length, contradicting M''s maximality. ∎

1.8.9 Proposition

Let M be fg over a Notherian ring A.

(1) If $M \neq 0$, then M has a submodules $N \subseteq M$ such that $N \cong A/\mathfrak{p}$ for some prime $\mathfrak{p} \leq A$.

(2) There exists a chain $0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ of submodules such that for each i, there is an isomorphism of A-modules $M_{i+1}/M_i \cong A/\mathfrak{p}_i$ for prime ideals $\mathfrak{p}_i \leq A$.

(3) The chain in the previous point is not necessarily unique, but

- **(i)** for a prime ideal $\mathfrak{p} \leq A$ we have ann$M \subseteq \mathfrak{p}$ iff $\mathfrak{p}_i \subseteq \mathfrak{p}$ for some i. In particular, ann$M \subseteq \mathfrak{p}_i$ for all i, and the minimal ideals between $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ are precisely minimal prime ideals ann$(M) \subseteq \mathfrak{p}$.
- **(ii)** for every minimal prime ideal ann$(M) \subseteq \mathfrak{p}$, the $\mathcal{A}_\mathfrak{p}$ module $M_\mathfrak{p}$ has finite length equal to the number of times $\mathfrak{p}$ occurs in $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$.

Proof

Let Θ be the set of all ideal $I \leq A$ of the form $I = \text{ann}(m)$ for $0 \neq m \in M$. Since A is Noetherian, there is an m such that ann(m) is maximal. Since $\langle m \rangle \subseteq M$ is isomorphic to $A/\text{ann}(m)$, it is sufficient to show ann(m) is prime. So suppose $a, b \notin \text{ann}(m)$, then $bm \neq 0$ and ann$(m) \subseteq$ ann(bm). Thus by maximality ann$(m) =$ ann(bm) and then $a \notin \text{ann}(bm)$ so $abm \neq 0$ meaning $ab \notin \text{ann}(m)$, showing it is prime. This proves the first point.

Let Σ be the set of all submodules of M which possess such a chain. It is nonempty since the zero module is in it. Since M is Noetherian, let M' be a maximal element of Σ.

Now consider $M'' = M/M'$; if $M'' = 0$ we are finished. Otherwise by the first point it has a submodule $N'' = N'/M'$ isomorphic to $A/\mathfrak{p}$. But then composing the chain of M' with $\subsetneq N'$ gives such a chain, contradicting M''s maximality.

For the final point, notice that for $M' \subsetneq M$,

$$\text{ann}(M') \cdot \text{ann}(M/M') \subseteq \text{ann}(M) \subseteq \text{ann}(M') \cap \text{ann}(M/M').$$

The second inclusion is clear, the first is because for $a \in \operatorname{ann}(M')$ and $b \in \operatorname{ann}(M/M')$ and $m \in M$ then $bm \in M'$ and $abm \in aM' = 0$. Thus for a prime ideal $\mathfrak{p} \leq A$, $\operatorname{ann}(M) \subseteq \mathfrak{p}$ iff $\operatorname{ann}(M') \subseteq \mathfrak{p}$ or $\operatorname{ann}(M/M') \subseteq \mathfrak{p}$. Thus by induction $\operatorname{ann}(M) \subseteq \mathfrak{p}$ iff $\operatorname{ann}(M_{i+1}/M_i) \subseteq \mathfrak{p}$ for some i. Since $\operatorname{ann}(A/\mathfrak{p}_i) = \mathfrak{p}_i$, the first point follows.

For the second point, the chain $\{M_i\}$ gives the chain $\left\{S^{-1}M_i\right\}$ and

$$S^{-1}M_{i+1}/S^{-1}M_i \cong S^{-1}(M_{i+1}/M_i) \cong S^{-1}(A/\mathfrak{p}_i) \cong S^{-1}A/S^{-1}\mathfrak{p}_i.$$

Moreover $S^{-1}\mathfrak{p}_i = S^{-1}A$ iff $S \cap \mathfrak{p}_i \neq \emptyset$, and otherwise is a prime ideal. For $S = A - \mathfrak{p}$ for a minimal prime ideal $\operatorname{ann}(M) \subseteq \mathfrak{p}$, then $S^{-1}\mathfrak{p}_i = A_\mathfrak{p}$ if $\mathfrak{p} \neq \mathfrak{p}_i$ and otherwise $S^{-1}\mathfrak{p}_i = \mathfrak{p}A_\mathfrak{p}$ is a maximal ideal. Thus the sequence $(M_i)_\mathfrak{p}$ forms a composition series after deleting the repeated entries (when $\mathfrak{p} \neq \mathfrak{p}_i$), and we are left with a composition series (since the quotients are quotients by a maximal ideal) whose length is equal to the number of $\mathfrak{p}_i$ equal to $\mathfrak{p}$. ■

1.8.10 Definition

A minimal prime ideal is a prime ideal not containing any other prime ideal.

1.8.11 Lemma

A Noetherian ring A has finitely many minimal prime ideals.

Proof

The minimal prime ideals of A are the minimal primes containing $\operatorname{ann}(A) = 0$. Then for any chain as in the above lemma (whose components are $A/\mathfrak{p}$ for some prime), the minimal prime ideals are the minimal primes from $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$ which is finite. ■

1.8.12 Proposition

A ring is Artinian iff it is Noetherian and every prime ideal is maximal.

Proof

If A is Noetherian and every prime ideal is maximal, then every prime ideal is also minimal. Thus A has finitely many prime/maximal ideals, $\mathfrak{m}_1, \ldots, \mathfrak{m}_k$. Then their product $I = \mathfrak{m}_1 \cdots \mathfrak{m}_k$ is contained in the intersection of all prime ideals, $\operatorname{nil}(A)$. Because A is Noetherian I is finitely generated, so $I^n = 0$ for some n (since every element of I is nilpotent, and it is finitely generated).

We have a chain of submodules

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_k = A,$$

where $M_{i+1}/M_i \cong A/\mathfrak{m}_i$ for some i (by the above proposition). As a subquotient, M_{i+1}/M_i is a Noetherian A-module, and thus a Notherian module over the field $K = A/\mathfrak{m}_i$. A Noetherian vector space is finite-dimensional (it is fg), so that M_{i+1}/M_i is of finite length over K (its length is equal to its dimension). Then M_{i+1}/M_i has finite length over A as well, and then inductively we see that each M_i has finite length, and in particular A is Artinian.

Conversely if A is Artinian, then all of its prime ideals are maximal (homework), so we need only show it is Noetherian. If $\mathfrak{m}_1, \ldots, \mathfrak{m}_{n+1}$ are distinct maximal ideals then $\mathfrak{m}_1 \cdots \mathfrak{m}_n \supset \mathfrak{m}_1 \cdots \mathfrak{m}_{n+1}$ is strict, so by Artinian-ness there must be finitely many maximal ideals. If the inclusion were not strict then $\mathfrak{m}_1 \cdots \mathfrak{m}_n \subseteq \mathfrak{m}_{n+1}$, and then by $\mathfrak{m}_{n+1}$'s primality, $\mathfrak{m}_i \subseteq \mathfrak{m}_{n+1}$ for some i. But then $\mathfrak{m}_i = \mathfrak{m}_{n+1}$ by maximality, a contradiction to distinctness.

So let $I = \mathfrak{m}_1 \cdots \mathfrak{m}_n$ be the product of all A's prime/maximal ideals. Consider the descending chain $I \supseteq I^2 \supseteq I^3 \supseteq \cdots$ which must terminate, so $I^k = I^{k+1}$ eventually. We claim that $I^k = 0$.

If not, let $x_1, \ldots, x_{k+1} \in I$ such that $x_1 \cdots x_{k+1} \neq 0$. Because $I^k = I^{k+1}$, $\prod_{i=2}^{k+1} x_i = \sum_\alpha y_{\alpha,1} \cdots y_{\alpha,k+1}$ where the sum is over a finite set and each $y_{\alpha,i} \in I$. Thus $x_1 y_{\alpha_0,1} \cdots y_{\alpha_0,k+1} \neq 0$ for some α_0. Then from the original tuple $x_1, \ldots, x_{k+1} \in I$ we obtain a new tuple $x_1' = x_1, x_2', \ldots, x_{k+2}'$ ($x_i' = y_{\alpha_0,i-1}$) with nonzero product, and the first element unchanged.

We similarly may write the product of the final k elements $x_3' \cdots x_{k+2}'$ as an element of I^{k+1} and obtain $x_1, x_2', x_3', x_4'', \ldots, x_{k+3}''$ with nonzero product. Continuing this inductively, we produce an infinite sequence $y_1, y_2, \ldots$ such that $y_1 \cdots y_m \neq 0$ for all $m > 0$. Now consider the descending chain $(y_1) \supseteq (y_1 y_2) \supseteq \cdots$, which stabilizes since A is Artinian. Thus for some $m > 0$ and $a \in A$,

$$\prod_{i=1}^m y_i = a \prod_{i=1}^{m+1} y_i \implies \left(\prod_{i=1}^m y_i\right)(1 - a y_{m+1}) = 0.$$

Since $y_{m+1} \in I$ and thus $a y_{m+1} \in I$ is in the Jacobson radical (I is the product of all maximal ideals, and thus contained in their intersection), we have that $1 - a y_{m+1} \in A^\times$. Thus $\prod_{i=1}^m y_i = 0$, a contradiction.

So $I^k = 0$, and so there is a chain (taking $M_j = \mathfrak{m}_1 \cdots \mathfrak{m}_{n-j}$),

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_m = A$$

where M_{i+1}/M_i is annihlated by some $\mathfrak{m}_i$, and thus is Artinian over the field $K = A/\mathfrak{m}_i$ so finite-dimensional over A. Thus M_{i+1}/M_i has finite length over $A/\mathfrak{m}_i$, and thus over A, and then by induction we see that each M_i has finite length, in particular so does A and thus is Noetherian. ■

Note:

Recall that $\mathbb{Z}$ is Notherian but not Artinian. This is because while all of its non-zero prime ideals are maximal, (0) is a prime ideal which is not maximal. We see that in general an integral domain which is not a field cannot be Artinian.

1.9 Tensor products

I assume some familiarity with the tensor product (see e.g. my summary on representation theory otherwise). The tensor product is a bifunctor

$$\bullet \otimes_B \bullet \colon {}_A\mathrm{Mod}_B \times {}_B\mathrm{Mod}_C \to {}_A\mathrm{Mod}_C,$$

with the universal property that it represents the balanced map functor:

$$\hom(M \otimes_B N, P) \cong \mathrm{Bal}(M, N;\, P).$$

Balanced maps are maps $M \times N \to P$ which are additive and multiplicative in each coordinate, and $f(mb, n) = f(m, bn)$ for all $b \in B$.

The tensor product also has a right-adjoint. For an (A, B)-bimodule M, $\bullet \otimes_A M$ is a functor $\mathrm{Mod}_A \to \mathrm{Mod}_B$ with right adjoint $\hom_B(M, \bullet)$.

In the case of commutative rings (this course), all modules are bimodules, so the tensor product is a bifunctor

$$\bullet \otimes_A \bullet \colon \mathrm{Mod}_A \to \mathrm{Mod}_A,$$

which represents the bilinear map functor (bilinear in the usual sense).

Note still that we use tensor products between module categories. In particular if B is an A-algebra, then $B \otimes_A M$ is a B-module ($b(c \otimes m) = (bc) \otimes m$). This defines a map $\mathrm{Mod}_A \to \mathrm{Mod}_B$, and its right-adjoint is the restriction map (restrict a B-module to an A-module). Note that this is a special case of the tensor-hom adjunction: $\bullet \otimes_A B$ is left-adjoint to $\hom_B(B, \bullet) \cong B$.

So we have a natural isomorphism

$$\hom_B(B \otimes_A M, P) \cong \hom_A(M, P|_A), \qquad M \in \mathrm{Mod}_A, P \in \mathrm{Mod}_B.$$

The correspondence is given by mapping $f \colon B \otimes_A M \to P$ to $\tilde{f} \colon M \to P$ by $\tilde{f}(m) = f(1 \otimes m)$. And $g \colon M \to P|_A$ to $\hat{g} \colon B \otimes_A M \to P$ by $\hat{g}(1 \otimes m) = g(m)$.

1.9.1 Proposition

Let M be an A-module.

(1) For a multiplicative set $S \subseteq A$, $S^{-1}A \otimes_A M$ is naturally isomorphic to $S^{-1}M$.

(2) For an ideal $I \leq A$, $A/I \otimes_A M$ is naturally isomorphic to M/IM.

Proof

The canonical map $M \to S^{-1}M$ mapping $m \mapsto m/1$ defines a map $f\colon S^{-1}A \otimes_A M \to S^{-1}M$ by $f(1 \otimes m) = m/1$. Then

$$f(a/s \otimes m) = a/sf(1 \otimes m) = am/s.$$

We define its inverse $g\colon S^{-1}M \to S^{-1}A \otimes_A M$ by

$$g(m/s) = 1/s \otimes m.$$

We claim this is well-defined: if $m/s = m'/s'$ then $1/s \otimes m = 1/s' \otimes m'$. If $t(ms' - m's)$ then

$$\frac{1}{s} \otimes m = \frac{ts'}{ts's} \otimes m = \frac{1}{ts's} \otimes ts'm = \frac{1}{ts's} \otimes tsm' = \frac{1}{s'} \otimes m'$$

as desired, and clearly g is a module morphism. f and g are also clearly morphisms.
The proof for the second point is similar: projection $M \to M/IM$ is an A-module morphism. Thus this gives rise to an A/IA-module morphism $f\colon (A/IA) \otimes_A M \to M/IM$ by $f(1 \otimes m) = \overline{m}$. Then

$$f(\overline{a} \otimes m) = af(1 \otimes m) = a\overline{m} = \overline{am}.$$

Its inverse is $g\colon M/IM \to (A/IA) \otimes_A M$ given by $g(\overline{m}) = 1 \otimes m$; this is easily seen to be well-defined and an inverse. ∎

1.9.2 Corollary

Let B be an A-algebra and M a B-module. Then there is a natural map

$$B \otimes_A M|_A \to M, \qquad b \otimes m \to bm.$$

In particular this is a natural map $m_B\colon B \otimes_A B \to B$ by $b \otimes b' \mapsto bb'$.

Proof

This map is the map corresponding to the identity of $\hom_A(M|_A, M|_A)$. The second point is for $M = B$. ∎

1.9.3 Lemma

If B, C are A-algebras, then their tensor product $B \otimes_A C$ is naturally an A-algebra and their coproduct in the category of A-algebras.

Proof

Let $D = B \otimes_A C$. The natural algebra structure is given by $(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$, and $A \to D$ by $a \mapsto a(1 \otimes 1)$. This is the composition of

$$D \times D \to D \otimes_A D, \qquad (d, d') \mapsto d \otimes d',$$

and

$$D \otimes_A D = (B \otimes_A C) \otimes_A (B \otimes_A C) \cong (B \otimes_A B) \otimes_A (C \otimes_A C) \to B \otimes_A C$$

where the final map is the tensor of $m_B \otimes m_C$.
The inclusion maps $i_B \colon B \to D$ and $i_C \colon C \to D$ are given by $b \mapsto b \otimes 1$ and $c \mapsto 1 \otimes c$.
Now if E is another A-algebra and there are maps $f \colon B \to E, g \colon C \to E$, then if $h \colon D \to E$ commutes with the rest of the diagram, i.e.

$$g = h \circ i_C, \qquad f = h \circ i_B,$$

then we must have

$$h(b \otimes c) = h((b \otimes 1)(1 \otimes c)) = h(b \otimes 1)h(1 \otimes c) = hi_B(b)hi_C(c) = f(b)g(c).$$

Thus h is unique, and this is well-defined (it is bilinear). ■

Recall the notion of left- and right-exact functors (preserving finite limits and colimits respectively). Tensor products are left-adjoints and thus right-exact:

1.9.4 Proposition

If $M' \to M \to M'' \to 0$ is an exact sequence of A-modules, then for any A-module N, the induced sequence

$$M' \otimes_A N \to M \otimes_A N \to M'' \otimes_A N \to 0$$

is exact.

Note:

Tensor products are not exact; they do not preserve injections. Indeed, if $f \colon A \to A$ is the map $f(x) = ax$ for $a \in A$, then $f \otimes M = f \otimes \mathrm{id}_M$ is the endomorphism of $A \otimes_A M \cong M$ which maps $m \mapsto am$. When a is not a zero divisor (e.g. A is a domain), f is injective, but $f \otimes M$ need not be (e.g. $M = A/(a)$).

Since $\hom_A(M, \bullet)$ is right-adjoint, it is left exact so it preserves finite limits. It is not hard to show that $\hom_A(\bullet, M)$ is left-exact as well (but contravariantly). Furthermore the Yoneda embedding reflects exactness:

1.9.5 Proposition

A sequence of A-modules $M_\bullet = M' \to M \to M'' \to 0$ is exact iff

$$\hom(M_\bullet, N) = 0 \to \hom(M'', N) \to \hom(M, N) \to \hom(M', N)$$

is exact for all $N \in \mathrm{Mod}_A$. Similarly $N_\bullet = 0 \to N' \to N \to N''$ is exact iff

$$\hom(M, N_\bullet) = 0 \to \hom(M, N') \to \hom(M, N) \to \hom(M, N'')$$

is exact for all M.

Proof

The "only if" part is due to left exactness. The converse (second point) is done by choosing $M = N'$ and looking at where id goes (this shows that the composition of maps is zero), then M to be the kernel of the second map, and showing that this implies exactness at M, and then you just need to show the first map is injective. ■

While not all tensor products are exact, some are. We give this case a special term.

1.9.6 Definition

An A-module M is **flat** if the functor $\bullet \otimes_A M$ is exact.

Note:

Recall that a functor is exact iff it preserves finite limits and colimits iff it preserves short exact sequences iff it preserves arbitrary exact sequences.

Since $\bullet \otimes_A M$ is right-exact for any module M, M is flat iff $\bullet \otimes_A M$ preserves injections.

1.9.7 Example

If $S \subseteq A$ is multiplicative, then $\bullet \otimes_A S^{-1}A$ is naturally isomorphic to $S^{-1}\bullet$, which is exact. Thus $S^{-1}A$ is a flat A-module.

If M is free, so $M \cong A^{\oplus I}$, then

$$\bullet \otimes_A M \cong \bullet \otimes_A A^{\oplus I}(\bullet \otimes_A A)^{\oplus I} \cong \bullet^{\oplus I}.$$

The second isomorphism is due to left adjoints preserving colimits. And $\bullet^{\oplus I}$ is exact (it maps injective maps to injective maps, more generally direct sums in module categories are exact, satisfying Grothendieck's axiom (AB4)).

1.9.8 Lemma

Let M be an A-module and $S \subseteq A$ a multiplicative set.

(1) If N is an $S^{-1}A$-module, it is an A-module via the canonical map $A \to S^{-1}A$, and there is a natural isomorphism

$$M \otimes_A N \to S^{-1}M \otimes_{S^{-1}A} N, \qquad m \otimes n \mapsto m/1 \otimes n.$$

(2) For an A-module N, there is a natural isomorphism

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \to S^{-1}(M \otimes_A N), \qquad \frac{m}{s} \otimes \frac{n}{t} \mapsto \frac{m \otimes n}{st}.$$

In particular for every prime $\mathfrak{p} \leq A$, there is a natural isomorphism

$$M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} \cong (M \otimes_A N)_{\mathfrak{p}}.$$

Proof

There is a natural isomorphism (by associativity of the tensor)

$$S^{-1}M \otimes_{S^{-1}A} N \cong (M \otimes_A S^{-1}A) \otimes_{S^{-1}A} N \cong M \otimes_A N$$

which maps $m/s \otimes n$ to $m \otimes 1/s \otimes n$ to $m \otimes n/s$. So in reverse it maps $m \otimes n$ to $m/1 \otimes n$ as desired.
There is a natural isomorphism

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \cong M \otimes_A S^{-1}N \cong M \otimes_A (N \otimes_A S^{-1}A) \cong (M \otimes_A N) \otimes_A S^{-1}A,$$

and this is naturally isomorphic to $S^{-1}(M \otimes_A N)$. ■

1.9.9 Corollary

Flatnes is local, i.e. the following are equivalent:

(1) M is a flat A-module,

(2) $S^{-1}M$ is a flat $S^{-1}A$-module for all multiplicative $S \subseteq A$,

(3) $M_\mathfrak{p}$ is a flat $A_\mathfrak{p}$-module for all prime $\mathfrak{p} \leq A$,

(4) $M_\mathfrak{m}$ is a flat $A_\mathfrak{m}$-module for all maximal $\mathfrak{m} \leq A$.

Proof

If M is flat then by the above lemma

$$\bullet \otimes_{S^{-1}A} S^{-1}M \cong S^{-1}(\bullet \otimes_A M),$$

which is a composition of exact functors and thus exact.
Now if $M_\mathfrak{m}$ is flat for all maximal $\mathfrak{m} \leq A$, we show M is flat. Recall we need only show that $\bullet \otimes_A M$ preserves injectives. Recall that f is injective iff $f_\mathfrak{m}$ is injective for all maximal $\mathfrak{m} \leq A$. Now, $f_\mathfrak{m}$ is isomorphic to $f \otimes M_\mathfrak{m}$, which is injective since $M_\mathfrak{m}$ is flat, we see $f_\mathfrak{m}$ is injective for all $\mathfrak{m}$ as desired. ■

I. Algebraic Geometry

1 Affine Schemes

1.1 Basic definitions

1.1.1 Definition

Let R be a ring, then its **(prime) spectrum** is its set of prime ideals,

$$\operatorname{Spec}(R) = \{\mathfrak{p} \leq R \mid \mathfrak{p} \text{ is prime}\}.$$

Note:

We can similarly define the **maximal spectrum** of a ring to be its set of maximal ideals. This is in general less useful than the prime spectrum, but its definition is analogous:

$$\operatorname{Specm}(R) = \{\mathfrak{m} \leq R \mid \mathfrak{m} \text{ is maximal}\}.$$

We can give $\operatorname{Spec}(R)$ a topological structure. Instead of defining the open sets of a spectrum, we will define its closed sets.

1.1.2 Definition

Let $S \subseteq R$ be a subset, then define its **vanishing set** to be

$$V(S) = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid S \subseteq \mathfrak{p}\}.$$

Notice that clearly we have

$$V(S) = V((S)),$$

so that every vanishing set is a vanishing set of an ideal. Note that $V(I) = V(J)$ does not imply $I = J$: for instance $I = 2\mathbb{Z}$ and $J = 4\mathbb{Z}$ in $R = \mathbb{Z}$; their vanishing sets are both $\{2\mathbb{Z}\}$.

1.1.3 Lemma

A proper ideal $\mathfrak{p} \leq R$ is prime iff $IJ \leq \mathfrak{p}$ implies $I \leq \mathfrak{p}$ or $J \leq \mathfrak{p}$ for all ideals $I, J \leq R$.

Proof

The "if" part is clear: if $ab \in \mathfrak{p}$, then $(a) \cdot (b) = (ab) \in \mathfrak{p}$ so either (a) or (b) is a subset of $\mathfrak{p}$. Conversely, suppose that $\mathfrak{p}$ is a prime ideal and $IJ \leq \mathfrak{p}$ but I, J are both not subsets of $\mathfrak{p}$. Then let $x \in I, y \in J$ be elements not in $\mathfrak{p}$: we then have $xy \in IJ \leq \mathfrak{p}$, a contradiction to $\mathfrak{p}$'s primality. ■

1.1.4 Proposition

The collection of vanishing sets forms the collection of closed sets of a topology on $\operatorname{Spec}(R)$.

Proof

We need to show that $\operatorname{Spec}(R), \emptyset$ are vanishing sets, the arbitrary intersection of vanishing sets is a vanishing set, and the binary union of vanishing sets is a vanishing set. Clearly we have

$$\operatorname{Spec}(R) = V(0), \qquad \emptyset = V(R)$$

so the first point is true.
Now consider a collection $\{S_\alpha\}_\alpha$ of subsets. Then we have:

$$\bigcap_\alpha V(S_\alpha) = V\left(\bigcup_\alpha S_\alpha\right),$$

since $S_\alpha \leq \mathfrak{p}$ for all α iff $\bigcup_\alpha S_\alpha \leq \mathfrak{p}$. In terms of ideals, we have that

$$\bigcap_\alpha V(I_\alpha) = V\left(\left(\bigcup_\alpha I_\alpha\right)\right) = V\left(\sum_\alpha I_\alpha\right).$$

For the final point on binary unions, we can restrict our attention to vanishing sets of ideals. Due to the above lemma we have that

$$V(I) \cup V(J) = V(IJ).$$

Thus the collection of vanishing sets forms a topology of closed sets. ■

1.1.5 Definition

The topology defined above on $\operatorname{Spec}(R)$ is called the **Zariski topology** of $\operatorname{Spec}(R)$.

1.1.6 Example

Consider $R = \mathbb{C}[x]$, a PID. Its prime ideals are the principal ideals generated by irreducible elements and zero, in this case

$$\operatorname{Spec}(\mathbb{C}[x]) = \{(x - \lambda) \mid \lambda \in \mathbb{C}\} \cup \{(0)\}.$$

For a polynomial $f \in \mathbb{C}[x]$, we have that $(x - \lambda) \in V(f)$ iff λ is a root of f. Thus

$$V(f) = \{(x - \lambda) \mid f(\lambda) = 0\}.$$

As we showed above, for $I \leq \mathbb{C}[x]$, we have that

$$V(I) = V\left(\bigcup_{f \in I} \{f\}\right) = \bigcap_{f \in I} V(f) = \{(x - \lambda) \mid \lambda \text{ is a root of every polynomial in } I\}.$$

Thus $\operatorname{Spec}(\mathbb{C}[x])$'s proper closed sets are precisely the finite subsets of $\operatorname{Spec}(\mathbb{C}[x]) - (0)$, i.e. it induces the cofinite topology on this subspace. The ideal (0) is not contained in any proper closed set, so its closure is all of $\operatorname{Spec}(\mathbb{C}[x])$.

1.1.7 Example

Consider $R = \mathbb{Z}$, a PID. Its prime ideals are the principal ideals generated by primes and zero, so

$$\operatorname{Spec}(\mathbb{Z}) = \{(p) \mid p \text{ prime}\} \cup \{(0)\}.$$

For $n \in \mathbb{Z}$, $(p) \in V(n)$ iff p divides n so that

$$V(n) = \{(p) \mid p \text{ is a prime divisor of } n\}.$$

Thus $\mathrm{Spec}(\mathbb{Z})$'s proper closed sets are precisely the finite subsets of $\mathrm{Spec}(\mathbb{Z}) - 0$, so it again induces the cofinite topology on this subspace. As before, (0)'s closure is all of $\mathrm{Spec}(\mathbb{Z})$.

In the two examples above, we saw that the point (0)'s closure is all of $\mathrm{Spec}(R)$. If (0) is a prime ideal, then this is always true: $(0) \in V(I)$ iff $I = (0)$ in which case $V(0) = \mathrm{Spec}(R)$. Thus $\mathrm{Spec}(R)$ is the only closed set containing the zero ideal, and is thus its closure.

1.1.8 Example

Consider the case that $R = k$ is a field. The only prime ideal of k is the zero ideal, so $\mathrm{Spec}(k)$ is a single-point space. There is only one topology on such a space: the trivial and discrete topology coincide.

The map $R \mapsto \mathrm{Spec}(R)$ is functorial.

1.1.9 Definition

Let $\varphi\colon R \to S$ be a ring morphism, then define $\varphi^*\colon \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ by

$$\varphi^*(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q}), \qquad \mathfrak{q} \in \mathrm{Spec}(S).$$

This is well-defined, as we showed before that the preimage of a prime ideal is prime.

1.1.10 Proposition

φ^* is continuous, and functorial.

Proof

To show that φ^* is continuous, we need to show that its preimage on closed sets is closed. So let $I \leq R$ be an ideal, and consider $(\varphi^*)^{-1}V(I)$:

$$\mathfrak{q} \in (\varphi^*)^{-1}V(I) \iff \varphi^{-1}\mathfrak{q} \in V(I) \iff I \leq \varphi^{-1}\mathfrak{q} \iff \varphi(I) \leq \mathfrak{q},$$

so that we have

$$(\varphi^*)^{-1}V(I) = V(\varphi(I)),$$

which is closed as desired. Note that while $\varphi(I) \subseteq S$ may not be an ideal, all vanishing sets are closed. Functorality is clear. ∎

Note:
As $\mathrm{Specm}(R) \subseteq \mathrm{Spec}(R)$, we endow $\mathrm{Specm}(R)$ with the subspace topology. In particular its closed sets are

$$V_m(S) = \{\mathfrak{m} \in \mathrm{Specm}(R) \mid S \subseteq \mathfrak{m}\}.$$

But note that $R \mapsto \mathrm{Specm}(R)$ is not functorial: the above φ^* does not necessarily restrict to maximal ideals.

Let us consider more the Zariski topology. Clearly $V(\bullet)$ is contravariant (order reversing).

1.1.11 Proposition

Let $I, J \leq R$ be ideals, then $V(I) = V(J)$ iff $\sqrt{I} = \sqrt{J}$. In particular $V(I) = V(\sqrt{I})$ and $\mathrm{Spec}(R) = V(\mathrm{nil}(R))$.

Proof

We know that $\sqrt{I}$ is the intersection of all prime ideals containing I, meaning

$$\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}.$$

So the first implication is clear: if $V(I) = V(J)$ then $\sqrt{I} = \sqrt{J}$. Now if $\sqrt{I} = \sqrt{J}$, let $\mathfrak{p} \in V(I)$. If $\mathfrak{p} \notin V(J)$ then there is an $f \in J$ not in $\mathfrak{p}$, but then $f \in \sqrt{J}$ and $f \notin \mathfrak{p}$ so $f \notin \sqrt{I}$, a contradiction. ■

Note:
Let the **Jacobson radical** of an ideal $I \leq R$ be the intersection of all maximal ideals containing I:

$$\mathrm{Jac}_R(I) = \bigcap_{I \leq \mathfrak{m}} \mathfrak{m} = \bigcap_{\mathfrak{m} \in V_m(I)} \mathfrak{m}.$$

We can alter the proof above to show that $V_m(I) = V_m(J)$ iff $\mathrm{Jac}_R(I) = \mathrm{Jac}_R(J)$. Thus $V_m(I) = V_m(\mathrm{Jac}_R(I))$ and $\mathrm{Specm}(R) = V_m(\mathrm{Jac}(R))$, where $\mathrm{Jac}(R) = \mathrm{Jac}_R(R)$ is the usual Jacobson radical of a ring.

1.1.12 Definition

Let $Z \subseteq \mathrm{Spec}(R)$ be a subset. Define its **ideal** to be

$$\mathcal{I}(Z) = \{f \in R \mid Z \subseteq V(f)\}.$$

We can characterize $\mathcal{I}(Z)$ more concretely:

$$\mathcal{I}(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}.$$

Indeed, if $f \in \mathcal{I}(Z)$ then $Z \subseteq V(f)$ so for every $\mathfrak{p} \in Z$ we have that $\mathfrak{p} \in V(f)$, meaning $f \in \mathfrak{p}$. Conversely if $f \in \mathfrak{p}$ for every $\mathfrak{p} \in Z$, then $\mathfrak{p} \in V(f)$ for every $\mathfrak{p}$ and so $Z \subseteq V(f)$.
From this characterization it is obvious that $\mathcal{I}(\bullet)$ is also order reversing.

1.1.13 Proposition

(1) Let $Z \subseteq \mathrm{Spec}(R)$ be a subset, then $V(\mathcal{I}(Z)) = \overline{Z}$, Z's closure.

(2) Let $J \leq R$ be an ideal, then $\mathcal{I}(V(J)) = \sqrt{J}$.

Proof

For the first point we need to show that $V(\mathcal{I}(Z))$ is the smallest vanishing set containing Z. It contains Z since if $\mathfrak{p} \in Z$ then $\mathcal{I}(Z) \leq \mathfrak{p}$ by definition (it is the intersection of all such $\mathfrak{p}$), so $\mathfrak{p} \in V(\mathcal{I}(Z))$. Now if

$Z \subseteq V(J)$ then for $\mathfrak{p} \in Z$ we have that $J \leq \mathfrak{p}$, so that $J \subseteq \bigcap_{\mathfrak{p} \in Z} \mathfrak{p} = \mathcal{I}(Z)$. Since V is contravariant, we have that $V(\mathcal{I}(Z)) \subseteq V(J)$, as desired.

For the second point, we have

$$\mathcal{I}(V(J)) = \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \bigcap_{J \leq \mathfrak{p}} \mathfrak{p} = \sqrt{J},$$

as desired. ■

1.1.14 Corollary

Let $\mathfrak{p} \in \operatorname{Spec}(R)$ be a point, then its closure is $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$. In particular $\{\mathfrak{p}\}$ is closed iff $\mathfrak{p}$ is maximal.

Proof

This is because $\mathcal{I}\{\mathfrak{p}\} = \mathfrak{p}$, so apply the above proposition. And $V(\mathfrak{p}) = \{\mathfrak{p}\}$ iff $\mathfrak{p}$ is maximal. ■

1.1.15 Example

This shows that $\operatorname{Spec}(R)$ need not be T_1. Recall that a space X is T_1 if for every two distinct $x \neq y \in X$, there is an open set U such that $x \in U$ and $y \notin U$. Equivalently a space is T_1 iff its singletons are closed: if singletons are closed let $U = \{y\}^c$, and conversely if for every $y \neq x$ we choose a U_y such that $y \in U_y$ and $x \notin U_y$ we see that

$$X - \{x\} = \bigcup_{y \neq x} U_y$$

is open and thus $\{x\}$ is closed.

In any case, clearly if we have prime ideals which are not maximal, then $\operatorname{Spec}(R)$ is not T_1. For instance, take $X = \operatorname{Spec}(\mathbb{Z})$. Since $(0) \in X$ is not maximal, it is not closed and thus the space cannot be T_1.

$\operatorname{Spec}(R)$ *is* T_0 though: all distinct points are topologically distinct. Indeed, if $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec}(R)$ have the same neighborhoods they have the same closure and thus $V(\mathfrak{p}) = V(\mathfrak{q})$. Thus $\mathfrak{p} \in V(\mathfrak{q})$ and $\mathfrak{q} \in V(\mathfrak{p})$ so that $\mathfrak{p} \leq \mathfrak{q} \leq \mathfrak{p}$ so that they are equal.

1.1.16 Definition

Let P, Q be two posets. An **antitone Galois connection** are two order-reversing functions $F \colon P \to Q$ and $G \colon Q \to P$ such that for every $p \in P, q \in Q$,

$$q \leq Fp \iff p \leq Gq.$$

An alternative method of looking at antitone Galois connections is via category theory. Recall that we can view posets as categories, where there exists a unique morphism $a \to b$ iff $a \leq b$.

1.1.17 Proposition

An antitone Galois connection between two posets P, Q consists of two functors $F \colon P \to Q^{\mathrm{op}} \colon G$ such that $F \dashv G$ forms an adjunction.

Proof

Order reversing functions between posets are just contravariant functors. So if F, G form an antitone Galois connection, then they are functors $F\colon P^{\mathsf{op}} \to Q$ and $G\colon Q^{\mathsf{op}} \to P$. Then $F^{\mathsf{op}}\colon P \to Q^{\mathsf{op}}\colon G$ forms an adjunction: we have $q \leq Fp$ iff $F^{\mathsf{op}}p \leq^{\mathsf{op}} q$ iff $p \leq Gq$. The converse is shown similarly. ■

1.1.18 Corollary

$F\colon P \to Q\colon G$ forms an antitone Galois connection iff they are order reversing and $p \leq GFp$ and $q \leq FGq$ for all $p \in P, q \in Q$.

Proof

This is just the unit-counit formulation of an adjunction (remember contravariance!). A natural transformation between functors X, Y between posets amounts to saying $X(\bullet) \leq Y(\bullet)$. ■

1.1.19 Corollary

The two functions $V\colon \mathcal{P}(R) \to \mathcal{P}(\mathrm{Spec}(R))\colon \mathcal{I}$ form an antitone Galois connection.

Proof

The maps are order reversing and $S \subseteq \sqrt{(S)} = \mathcal{I}(V(S))$ and $Z \subseteq \overline{Z} = V(\mathcal{I}(Z))$, so they form an antitone Galois connection. Showing directly that $Z \subseteq V(S)$ iff $S \subseteq \mathcal{I}(Z)$ is easy to do as well. ■

1.2 Topological properties of affine schemes

1.2.1 Definition

Let R be a ring, $X = \mathrm{Spec}(R)$ its spectrum. For $f \in R$, define

$$X_f = X - V(f) = \{\mathfrak{p} \in \mathrm{Spec}(R) \mid f \notin \mathfrak{p}\}.$$

1.2.2 Proposition

The collection $\{X_f\}_{f \in R}$ forms a basis of open sets for $X = \mathrm{Spec}(R)$.

Proof

This is equivalent to showing that $\{V(f)\}_{f \in R}$ forms a basis of closed sets; that they are closed and every closed set is an intersection of closed basis sets. Clearly $V(f)$ are closed, and for $S \subseteq R$ we have

$$V(S) = V\left(\bigcup_{f \in S} \{f\}\right) = \bigcap_{f \in S} V(f)$$

as desired. ■

This shows that for an open set $X - V(S)$, we have

$$X - V(S) = \bigcup_{f \in S} X_f.$$

1.2.3 Proposition

Let R be a ring and $X = \operatorname{Spec}(R)$.

(1) Let $\{f_\alpha\}_\alpha$ be a collection of elements from R. Then $X = \bigcup_\alpha X_{f_\alpha}$ iff $(\{f_\alpha\}_\alpha) = R$.

(2) Let $f \in R$, then $X_f = X$ iff $f \in R^\times$.

(3) Let $f \in R$, then $X_f = \emptyset$ iff $f \in \operatorname{nil}(R)$.

Proof

We saw that

$$\bigcup_\alpha X_{f_\alpha} = X - V(\{f_\alpha\}_\alpha)$$

which is equal to X iff $V(\{f_\alpha\}_\alpha) = \emptyset$. This is iff

$$V(\{f_\alpha\}_\alpha) = V((\{f_\alpha\}_\alpha)) = V(R) \iff \sqrt{(\{f_\alpha\}_\alpha} = R.$$

Now $\sqrt{J} = R$ iff $J = R$, as $1 \in \sqrt{J}$ iff $1 \in J$. So this is iff $(\{f_\alpha\}_\alpha) = R$, as desired.
The second point follows from the first: $X_f = X$ iff $(f) = R$ iff $f \in R^\times$.
For the third point, $X_f = \emptyset$ iff $V(f) = X = V(0)$ iff $\sqrt{(f)} = \operatorname{nil}(R)$. This occurs iff $f \in \operatorname{nil}(R)$: clearly it implies this, and conversely $0 \leq (f) \leq \operatorname{nil}(R)$ and radicals are order-preserving so we get $\sqrt{(f)} = \operatorname{nil}(R)$. ∎

1.2.4 Corollary

$X = \operatorname{Spec}(R)$ is compact.

Proof

We need to show every open cover of X has a finite subcover. Let $\mathcal{U}$ be an open cover of X, then we can assume that $\mathcal{U}$ consists of basic open sets: $\mathcal{U} = \{X_{f_\alpha}\}_\alpha$. By the first point above, since $\mathcal{U}$ is a cover, $(\{f_\alpha\}_\alpha) = R$. Thus $1 \in (\{f_\alpha\}_\alpha)$ so there are finitely many $f_1, \dots, f_n$ such that $1 = x_1 f_1 + \cdots + x_n f_n$. Then $(f_1, \dots, f_n) = R$ and so again by the above proposition, $\{X_{f_1}, \dots, X_{f_n}\}$ is a finite subcover of $\mathcal{U}$, as desired. ∎

Note:
This result shows that the usual topological concept of compactness is in general not strong enough for algebraic geometry. Thus in algebraic geometry topological compactness is usually called **quasicompactness**, and compactness is reserved for a stronger concept.

1.2.5 Proposition

Let $f \in R$, and consider R_f, the localization of R at $\{f^n\}_{n<\omega}$. $\mathrm{Spec}(R_f)$ is homeomorphic to X_f.

Proof

Let $\iota: R \to R_f$ be the structure map. Then $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$ is a bijective correspondence between prime ideals of R_f and prime ideals of R which do not intersect $\{f^n\}$. A prime ideal $\mathfrak{p} \leq R$ does not intersect $\{f^n\}$ iff $f \notin \mathfrak{p}$. Thus $\iota^*: Y = \mathrm{Spec}(R_f) \to \mathrm{Spec}(R)$ is injective and its image are all prime ideals such that $f \notin \mathfrak{p}$, i.e. $\mathrm{im}\iota^* = X_f$.

We claim that ι^* is a homeomorphism onto its image. We already know it is continuous and bijective, so we need only show it is open; we will show it sends basic open sets to basic open sets. So let $g/f^n \in R_f$, with the basic open set

$$Y_{g/f^n} = \{\mathfrak{q} \in Y \mid g/f^n \notin \mathfrak{q}\}.$$

We claim that $\iota^*(Y_{g/f^n}) = X_g \cap X_f$. Indeed, if $\mathfrak{q} \in Y_{g/f^n}$ then $g/f^n \notin \mathfrak{q}$ and so $g \notin \iota^*\mathfrak{q}$ (since g/f^n is $\iota(g)$ times a unit, so if $g \in \iota^*\mathfrak{q}$ then $\iota(g) \in \mathfrak{q}$ and then $g/f^n \in \mathfrak{q}$); thus $\iota^*\mathfrak{q} \in X_g$ as desired. Conversely if $\mathfrak{p} \in X_f \cap X_g$ then $g/f^n \notin \mathfrak{p}R_f$: otherwise $g/f^n = h/f^m$ for $h \in \mathfrak{p}$ and thus $f^k(f^m g - f^n h) = 0$ so that $f^{m+k}g \in \mathfrak{p}$, but $f, g \notin \mathfrak{p}$ a contradiction. Now, $\mathfrak{p}R_f$ is prime and $\iota^*\mathfrak{p}R_f = \mathfrak{p}$, showing the equality. ■

We now focus on the connectedness of topological spaces, in particular spectra of rings. Recall that a topological space is **disconnected** if it can be written as the disjoint union of two open subsets, equivalently if it has a non-trivial clopen subset.

1.2.6 Lemma

A surjective ring morphism $\varphi: R \to S$ induces a closed embedding $\varphi^*: \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ (i.e. its image is closed, and it is a topological embedding). In particular if $I \leq R$ is an ideal, then the projection $\pi: R \to R/I$ induces a closed embedding $\pi^*: \mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$ whose image is $V(I)$.

Proof

φ induces an isomorphism of rings $R/\ker\varphi \to S$, which carries prime ideals of $R/\ker\varphi$ bijectively to prime ideals of S. Thus φ^*'s image is the set of all prime ideals containing $\ker\varphi$, i.e. $\mathrm{im}\varphi^* = V(\ker\varphi)$ which is closed by definition. Since φ is surjective, φ^* is injective and thus a bijection onto its image. So to show it is a topological embedding, we need only show that it is a closed map.

Indeed, for $J \leq S$,

$$\varphi^* V(J) = \left\{\varphi^{-1}\mathfrak{q} \,\middle|\, J \leq \mathfrak{q}\right\} = \left\{\varphi^{-1}\mathfrak{q} \,\middle|\, \varphi^{-1}J \leq \varphi^{-1}\mathfrak{q}\right\} = V(\varphi^{-1}J) \cap \mathrm{im}\varphi^*,$$

which is closed. (The second equality is due to φ's surjectivity.) ■

1.2.7 Proposition

If $R = S \times T$ is a product ring, then $\mathrm{Spec}(R)$ is homeomorphic to $\mathrm{Spec}(S) \sqcup \mathrm{Spec}(T)$, and thus disconnected. Conversely if $\mathrm{Spec}(R) = Y \sqcup Z$ is disconnected, then there exist rings S, T such that Y, Z are homeomorphic to $\mathrm{Spec}(S), \mathrm{Spec}(T)$ respectively, and $R \cong S \times T$.

Proof

Let $X = \mathrm{Spec}(R)$. Consider $(1,0), (0,1) \in R$, so that R's unit is their sum. Then we showed above that $X = X_{(1,0)} \cup X_{(0,1)}$, and this is a disjoint union since $V(0,1) \cup V(1,0) = X$ (if $\mathfrak{p} \in X$ and $(0,1), (1,0) \notin \mathfrak{p}$ then their product 0 is not in $\mathfrak{p}$, a contradiction). We also showed that $X_{(1,0)} \cong \mathrm{Spec}(R_{(1,0)})$.

So let $\pi\colon R \to S$ be the projection, so $\pi(1,0) = 1 \in S^\times$ so that by the universal property of localization π induces a map $\pi'\colon R_{(1,0)} \to S$ which is surjective since π is. Now if $\pi'((a,b)/(1,0)) = \pi'((x,y)/(1,0))$ then $a = x$ so $(a,b)(1,0) = (x,y)(1,0)$ and thus $(a,b)/(1,0) = (x,y)/(1,0)$ showing that π' is also injective. Thus $S \cong R_{(1,0)}$, and similarly $T \cong R_{(0,1)}$ so that we have desired result.

Since Y, Z are closed, there are $I, J \leq R$ such that $Y = V(I)$ and $Z = V(J)$. Then let $S = R/I$ and $T = R/J$; we showed above that $\pi_1\colon R \to R/I$ induces a homeomorphism $\pi_1^*\colon \mathrm{Spec}(R/I) \to V(I) \subseteq \mathrm{Spec}(R)$ and similar for R/J. Now, $Y \cap Z = V(I) \cap V(J) = V(I+J)$ is empty so that $I + J = R$, and thus I, J are coprime. By the chinese remainder theorem,

$$R \cong R/I \times R/J = S \times T,$$

as desired. ■

1.2.8 Definition

An element $f \in R$ is **idempotent** if $f^2 = f$. The **trivial idempotents** of R are 0 and 1.

If $f \in R$ is idempotent, then fR is a ring. Indeed, it is closed under addition and multiplication (it is the ideal generated by f), and f is its multiplicative identity since it is idempotent.

1.2.9 Proposition

$R \cong S \times T$ for non-trivial rings S, T iff R has nontrivial idempotents.

Proof

If $R \cong S \times T$ then $(1,0), (0,1)$ are nontrivial idempotents. Conversely let $f \in R$ be a nontrivial idempotent, then define $S = fR$ and $T = (1-f)R$. Then $R \to S \times T$ given by $g \mapsto (fg, (1-f)g)$ is clearly an injective since if $fg = (1-f)g = 0$ then $g = 0$, and it is surjective since for $(fg, (1-f)h) \in S \times T$ then we want a $x \in R$ such that $fg = fx$ and $(1-f)h = (1-f)x$, so $(1-f)h = x - fg$ and thus $x = (1-f)h + fg$ gives the desired element. They are also non-trivial since $f \neq 0$. ■

1.2.10 Definition

A ring R is **connected** if it has no nontrivial idempotents.

1.2.11 Corollary

$\mathrm{Spec}(R)$ is connected iff R is.

1.2.12 Definition

A topological space X is **reducible** if there exist two proper closed subsets $Z, W \subsetneq X$ such that $X = Z \cup W$. Otherwise X is **irreducible**.

Note that irreducibility implies connectedness; if X is disconnected then its union $Z \sqcup W$ of open sets are also closed. The converse is not true in general (take the axes of $\mathbb{R}^2$).

1.2.13 Proposition

$X = \operatorname{Spec}(R)$ is irreducible iff $\operatorname{nil}(R)$ is prime.

Proof

If $\operatorname{nil}(R)$ is not prime, then there are $f, g \notin \operatorname{nil}(R)$ such that $fg \in \operatorname{nil}(R)$. Let $Z = V(f)$ and $W = V(g)$, so that

$$Z \cup W = V(f) \cup V(g) = V(fg) = X,$$

where the second equality is because $V(f) \cup V(g) = V((f)(g)) = V(fg)$, and the final equality is because $V(fg) \supseteq V(\operatorname{nil}(R)) = X$.

Conversely if X is reducible, then $X = V(I) \cup V(J)$ so that $V(IJ) = X$ for $V(I), V(J) \neq X$. But then $IJ \leq \operatorname{nil}(R)$ and I and J cannot be contained in $\operatorname{nil}(R)$, so $\operatorname{nil}(R)$ is not prime. ■

1.2.14 Corollary

The spectrum of a domain is irreducible.

1.2.15 Example

The affine space $\mathbb{A}^n_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[x_1, \ldots, x_n])$ is irreducible since $\mathbb{C}[x_1, \ldots, x_n]$ is a domain.

Similarly the spectrum of $R = \mathbb{C}[x_1, \ldots, x_n]/(f)$ is irreducible when f is irreducible (and then (f) is prime and R is a domain). But the converse does not hold: let $R = \mathbb{C}[x]/(x^2)$, x^2 is reducible, and $\operatorname{nil}(R) = \sqrt{(x^2)} = (x)$ is prime.

1.3 Categorical constructions

In this section we consider the category of affine schemes, i.e. spectrums of rings. A morphism of affine schemes $\operatorname{Spec}(R) \to \operatorname{Spec}(S)$ corresponds to a ring morphism $S \to R$.

Recall the following case of a categorical limit.

1.3.1 Definition

A **pullback** (or **fiber product**) is the limit of a diagram of shape $\bullet \to \bullet \leftarrow \bullet$ (if one exists). Explicitly, given two morphisms $A \xrightarrow{f} C \xleftarrow{g} B$, their fiber product is an object $A \times_C B$ along with morphisms $A \xleftarrow{i} A \times_C B \xrightarrow{j} B$ such that the following diagram commutes and is universal (in the sense that it is the terminal such diagram which commutes):

$$\begin{tikzcd}
A \times_C B \arrow[r, "i"'] \arrow[d, "j"'] & A \arrow[d, "f"'] \\
B \arrow[r, "g"] & C
\end{tikzcd}$$

Finite products and equalizers can be written as pullbacks with a terminal object. In particular if 1 is a terminal object, then the pullback of $A \xrightarrow{!} 1 \xleftarrow{!} B$, where ! denotes the unique morphism, is the product of A and B. And if $f, g\colon A \rightrightarrows B$ are parallel morphisms, the pullback of $A \xrightarrow{(f,g)} B \times B \xleftarrow{\Delta} B$, where Δ is the diagonal morphism, is the equalizer of f, g.

Thus any category with pullbacks and a terminal object has all finite limits. We use this to demonstrate now that the category of affine schemes has all finite limits.

1.3.2 Proposition

Let $X = \operatorname{Spec}(A), Y = \operatorname{Spec}(B), Z = \operatorname{Spec}(R)$ be affine schemes, along with morphisms $X \to Z \leftarrow Y$, then this diagram has a pullback. Explicitly, its pullback is $X \times_Z Y = \operatorname{Spec}(A \otimes_R B)$.

Proof

The morphisms $X \to Z \leftarrow Y$ are induced by morphisms $A \leftarrow R \to B$, so A, B are R-algebras. Thus $A \otimes_R B$ is an R-algebra as well, and the obvious maps $A, B \to A \otimes_R B$ given by $a \mapsto a \otimes 1, b \mapsto 1 \otimes b$ give us the cone $X \times_Z Y \to X, Y$. We need to show that this is indeed a limit cone, i.e. that it is terminal.

We need to show that if $W = \operatorname{Spec}(C)$ is any other affine scheme with morphisms $W \to X, Y$ commuting with $X \to Z, Y \to Z$ then there is a unique morphism fro W to $X \times_Z Y = \operatorname{Spec}(A \otimes_R B)$. Translating this to the category of rings: if there are morphisms $A \xrightarrow{\varphi} C \xleftarrow{\psi} B$ which commute with $R \to A, B$, then there is a unique morphism $A \otimes_R B \to C$ making the diagram commute

$$\begin{tikzcd}
C & & \\
& A \otimes_R B \arrow[ul, dashed, "\exists!"] & A \arrow[l] \arrow[ull, bend right, "\varphi"'] \\
& B \arrow[u] \arrow[uul, bend left, "\psi"] & R \arrow[l] \arrow[u]
\end{tikzcd}$$

Such a map must map $a \otimes b$ to $\varphi(a) \otimes \psi(b)$. This map is R-bilinear (because φ, ψ commute with the maps $R \to A, B$), and thus is well-defined, as desired. ■

1.3.3 Corollary

The category of affine schemes has all finite limits.

Proof

The ring $\mathbb{Z}$ is initial: there is a unique ring morphism $\mathbb{Z} \to A$ for all rings A. Thus its spectrum is terminal, so the category of affine schemes has a terminal object. The previous proposition showed the category has all pullbacks, so it has all finite limits. ■

2 Varieties

2.1 Affine varieties

In this section let k be an algebraically closed field, $\operatorname{Fun}(k^n, k)$ denote the k-algebra of functions $k^n \to k$, and $k[x_1, \ldots, x_n]$ be the k-algebra of polynomials over variables $x_1, \ldots, x_n$.
Note that there is a natural k-algebra morphism

$$\operatorname{ev}\colon k[x_1, \ldots, x_n] \to \operatorname{Fun}(k^n, k)$$

defined by $\operatorname{ev}(f)(P) = p(a_1, \ldots, a_n)$ for a polynomial f and $P = (a_1, \ldots, a_n) \in k^n$.
For a subset $S \subseteq k[x_1, \ldots, x_n]$ define its **zero set**

$$Z(S) = \{P \in k^n \mid f(P) = 0 \text{ for all } f \in S\} \subseteq k^n.$$

For $f \in k^n$ we write $Z(f)$ for $Z(\{f\})$ so that $Z(S) = \bigcap_{f \in S} Z(f)$.

2.1.1 Lemma

(1) Z is order reversing: if $S_1 \subseteq S_2$ then $Z(S_1) \supseteq Z(S_2)$.

(2) $Z(S)$ is equal to $Z((S))$ ((S) is the ideal generated by S).

(3) For a collection of subsets $\{S_\alpha\}_\alpha$:

$$\bigcap_\alpha Z(S_\alpha) = Z\Big(\bigcup S_\alpha\Big).$$

(4) For two subsets S_1, S_2, define $S_1S_2 = \{fg \mid f \in S_1, g \in S_2\}$, then

$$Z(S_1) \cup Z(S_2) = Z(S_1S_2).$$

(5) $Z(1) = \emptyset$ and $Z(0) = k^n$.

Proof

All of these points are clear. We prove only the fourth: $Z(S_1) \cup Z(S_2) = Z(S_1S_2)$. Note that the ideal generated by S_1S_2 is precisely the product of the ideals generated by S_1 and S_2. Thus $Z(S_1)$ and $Z(S_2)$ are contained in $Z(S_1S_2)$ by order-reversion. Conversely if $P \notin Z(S_1) \cup Z(S_2)$, then $f(P), g(P) \neq 0$ for all $f \in S_1, g \in S_2$ so $(fg)(P) = f(P)g(P) \neq 0$ and thus $P \notin Z(S_1S_2)$. ■

2.1.2 Corollary

The set of zero sets forms a topology of closed sets on k^n.

2.1.3 Definition

Define $\mathbb{A}^n_k$ to be the topological space k^n whose underlying topology is given by the zero sets. This is called the **Zariski topology**, and $\mathbb{A}^n_k$ is the **n-dimensional affine space**. When k is understood, we omit it and just write $\mathbb{A}^n$.
We also define $k[\mathbb{A}^n] = k[x_1, \ldots, x_n]$.

Note that $Z(f)$ forms a basis of closed sets: every closed set is an intersection of these basic closed sets. Thus

$$D(f) = \mathbb{A}^n - Z(f) = \{P \in k^n \mid f(P) \neq 0\}.$$

forms a basis of open sets.
Note that for every $S \subseteq k[x_1, \ldots, x_n]$, (S) is generated by finitely many $f_1, \ldots, f_n \in S$ since $k[x_1, \ldots, x_n]$ is Noetherian. Then

$$Z(S) = Z((S)) = Z(f_1, \ldots, f_n) = \bigcap_{i=1}^n Z(f_i).$$

Thus every open set is a finite union of basic open sets. And so $\mathbb{A}^n$ is quasicompact: if $D(f_i)$ cover $\mathbb{A}^n$ then

$$\mathbb{A}^n = \bigcup_i D(f_i) = D(\{f_i\}),$$

and as we showed (for zero sets, thus for open sets) $D(\{f_i\})$ is the finite union of $D(f_i)$s.

2.1.4 Proposition

$\mathbb{A}^n$ is quasicompact.

2.1.5 Example

The closed subsets of $\mathbb{A}^1$ are precisely the finite subsets of $\mathbb{A}^1$, or $\mathbb{A}^1$ itself. Indeed, $k[x]$ is a PID so all of its ideals are of the form $Z(f)$ for a polynomial f. If $f \neq 0$, then f has finitely many roots so $Z(f)$ is finite. And for a finite subset $\{c_1, \dots, c_n\}$ of $\mathbb{A}^1 = k$, $f = (x - c_1) \cdots (x - c_n)$ has this set as its zero set.

Thus the open subsets of $\mathbb{A}^1$ are empty or cofinite. Since k is infinite (it is algebraically closed), $\mathbb{A}^1$ is not Hausdorff.

2.1.6 Definition

For a subset $Y \subseteq \mathbb{A}^n$, define

$$\mathcal{I}(Y) = \{f \in k[x_1, \dots, x_n] \mid f(P) = 0 \text{ for all } P \in Y\}.$$

2.1.7 Proposition

(1) $\mathcal{I}$ is order-reversing.

(2) $\mathcal{I}(Y)$ is a radical ideal of $k[x_1, \dots, x_n]$.

(3) $\mathcal{I}(Y_1 \cup Y_2) = \mathcal{I}(Y_1) \cap \mathcal{I}(Y_2)$.

(4) For every $Y \subseteq \mathbb{A}^n$, $Z(\mathcal{I}(Y)) = \overline{Y}$ is the closure of Y in $\mathbb{A}^n$.

(5) **Nullstellensatz**: for every ideal $J \leq k[x_1, \dots, x_n]$, $\mathcal{I}(Z(J)) = \sqrt{J}$.

Proof

The first three points are clear. For the fourth, notice $Y \subseteq Z(\mathcal{I}(Y))$ and $Z(\mathcal{I}(Y))$ is closed, so $\overline{Y} \subseteq Z(\mathcal{I}(Y))$. Since $\overline{Y}$ is closed, $\overline{Y} = Z(J)$ for some ideal J. So

$$Y \subseteq \overline{Y} = Z(J) \Longrightarrow \mathcal{I}(\overline{Y}) = \mathcal{I}(Z(J)) \subseteq \mathcal{I}(Y).$$

And $J \subseteq \mathcal{I}(Z(J))$ clearly, so $J \subseteq \mathcal{I}(Y)$, and thus $\overline{Y} = Z(J) \supseteq Z(\mathcal{I}(Y))$ as desired.

For the fifth point, since $J \leq \mathcal{I}(Z(J))$ which is radical, we see $\sqrt{J} \leq \mathcal{I}(Z(J))$. The inverse inclusion is far deeper, and we give a proof next section. ∎

2.1.8 Corollary

Z and $\mathcal{I}$ form an antitone Galois connection.

2.1.9 Corollary

For every proper ideal $J < k[x_1, \dots, x_n]$, $Z(J)$ is nonempty.

Proof

If $J \neq (1)$ then $\mathcal{I}(Z(J)) = \sqrt{J} \neq (1)$, thus $Z(J)$ cannot be empty (since $\mathcal{I}(\emptyset) = (1)$). ■

Note:
Recall that a topological space is irreducible iff there are no two proper closed subsets whose union is X. In homework this was shown to be equivalent to the condition that all open subsets intersect.

2.1.10 Example

Every closed subset of $\mathbb{A}^1$ is either $\mathbb{A}^1$ or finite. Since k is infinite, $\mathbb{A}^1$'s irreducible closed subsets are its closed points.

2.1.11 Corollary

A closed subset $Y \subseteq \mathbb{A}^n$ is irreducible iff the ideal $\mathcal{I}(Y) \subseteq k[\mathbb{A}^n]$ is either prime or all $k[\mathbb{A}^n]$.

Proof

If $Y \subseteq \mathbb{A}^n$ is irreducible and $fg \in \mathcal{I}(Y)$, then we want to show either $f \in \mathcal{I}(Y)$ or $g \in \mathcal{I}(Y)$. Notice that since $fg \in \mathcal{I}(Y)$, we have $Y = \overline{Y} = Z(\mathcal{I}(Y)) \subseteq Z(fg) = Z(f) \cup Z(g)$. Since Y is irreducible either $Y \subseteq Z(f)$ or $Y \subseteq Z(g)$, meaning $f \in \mathcal{I}(Y)$ or $g \in \mathcal{I}(Y)$ as desired.

If $\mathcal{I}(Y)$ is prime or all $k[\mathbb{A}^n]$, suppose $Y = Y_1 \cup Y_2$ where $Y_1, Y_2 \subseteq Y$ are closed. Then

$$\mathcal{I}(Y_1)\mathcal{I}(Y_2) \subseteq \mathcal{I}(Y_1) \cap \mathcal{I}(Y_2) = \mathcal{I}(Y).$$

Thus either $\mathcal{I}(Y_1)$ or $\mathcal{I}(Y_2)$ is contained in $\mathcal{I}(Y)$. Suppose $\mathcal{I}(Y_1) \subseteq \mathcal{I}(Y)$, then applying Z gives $Y_1 \supseteq Y$ (since they are closed), so $Y = Y_1$ as desired. ■

2.1.12 Example

$\mathbb{A}^n$ is irreducible since $\mathcal{I}(\mathbb{A}^n) = (0)$ is prime.

If $f \in k[\mathbb{A}^n]$ is irreducible, then $Z(f)$ is irreducible in $\mathbb{A}^n$. This is because $\mathcal{I}(Z(f)) = \sqrt{(f)} = (f)$ is prime.

Proof of the Nullstellensatz

2.1.13 Lemma

Let $f \in k[x_1, \dots, x_n]$ be nonconstant and $A = k[x_1, \dots, x_n]/(f)$. Then there exist integers $k_1, \dots, k_{n-1} > 0$ such that the k-algebra homomorphism

$$\varphi \colon k[y_1, \dots, y_{n-1}] \to A, \qquad y_i \mapsto x_i - x_n^{k_i}$$

is injective and finite.

Proof

For an n-tuple $I = (d_1, \dots, d_n)$ let $x^I = x_1^{d_1} \cdots x_n^{d_n}$ and call I the **multidegree** of x^I. We define k_j for $1 \le j \le n$ by descending recursion and setting $k_n = 1$, and letting

$$k_j = 1 + \max_I \left\{ \sum_{i=j+1}^n k_i d_i \right\},$$

where $I = (d_1, \dots, d_n)$ ranges over all the multidegrees in f with nonzero coefficient. We claim that k_j satisfy the claim.

Let $y_i = x_i - x_n^{k_i}$ for $1 \le i \le n-1$, and order the multidegrees in f lexicographically. That is, $(d_1, \dots, d_n) > (d'_1, \dots, d'_n)$ iff $d_j > d'_j$ for some index j and $d_i = d'_i$ for all $i < j$. This is a total order, and so f has a maximum multidegree $I_f = (D_1, \dots, D_n)$.

Now define $N = \sum_{i=1}^n k_i D_i$. Then we claim that f is a polynomial in $B[x_n]$ of degree $N > 0$ in x_n, whose leading coefficient is in $k^\times$, where $B = k[y_1, \dots, y_{n-1}] \subseteq k[x_1, \dots, x_n]$.

Let $m = x^I$ be a monomial with $I = (d_1, \dots, d_n)$. Substitute $x_i = y_i + x_n^{k_i}$, which shows that m is a monic polynomial in $B[x_n]$ of x_n-degree $\sum_{i=1}^n k_i d_i$. We claim that under this substitution, the x_n-degree of x^{I_f} is greater than x^I's for any other multidegree I in f. That is,

$$\sum_{i=1}^n k_i D_i > \sum_{i=1}^n k_i d_i$$

for all other multidegrees I. This will then imply the claim regarding f, as it is of the form

$$f = \sum_I a_I x^I = \sum_I a_I (y_I - x_n^{k_I})^I,$$

and considering the combination of monomials.

So let $I = (d_1, \dots, d_n)$ be another multidegree of f. By I_f's maximality there is a j such that $D_j \ge d_j + 1$ and $D_i = d_i$ for $i < j$. Then

$$\sum_{i=1}^n k_i D_i \ge \sum_{i=1}^j k_i D_i \ge \sum_{i=1}^{j-1} k_i d_i + k_j(d_j + 1) = \sum_{i=1}^j k_i d_i + k_j.$$

By definition of k_j,

$$> \sum_{i=1}^j k_i d_i + \sum_{i=j+1}^n k_i d_i = \sum_{i=1}^n k_i d_i$$

as desired.

Now we need to show that these k_i actually prove the lemma. First we show that φ is integral; let $C = k[y_1, \dots, y_{n-1}]$ be φ's domain. Let $c \in k^\times$ be f's leading coefficient as a polynomial in $B[x_n]$, then $c^{-1} f$ vanishes in A. Since $c^{-1} f$ is a monic polynomial in $B[x_n]$ which is zero, so $\overline{x}_n \in A$ is integral over C, as desired. Since $\overline{x}_i = \overline{y}_i + \overline{x}_n^{k_i}$, $\overline{x}_i \in A$ is also integral over C, and thus A is integral over C via φ. Furthermore, A is fg over C and thus finite.

Note that $x_n \mapsto x_n$, $x_i \mapsto y_i = x_i - x_n^{k_i}$ is invertible, and thus $k[x_1, \dots, x_n] = k[y_1, \dots, y_{n-1}][x_n]$, and $y_1, \dots, y_{n-1}, x_n$ are algebraically independent. Now suppose $\varphi(g) = 0$ for some $g \in C = k[t_1, \dots, t_{n-1}]$. That is, $g(\overline{y}_1, \dots, \overline{y}_{n-1}) = 0 \in A$ so $g(y_1, \dots, y_{n-1}) = hf$ for some $h \in k[y_1, \dots, y_{n-1}][x_n]$. Since g has zero in x_n, but f has positive degree, we must have that $g = 0$. So φ is injective. ■

2.1.14 Theorem (Noether normalization lemma)

Let $A \neq 0$ be a fg algebra over a field k. Then there exists a k-subalgebra $B \subseteq A$ such that B is isomorphic as a k-algebra to a polynomial algebra $k[y_1, \ldots, y_m]$ and A is a finite B-algebra.

Proof

Since A is fg, $A \cong k[x_1, \ldots, x_n]/I$ for some ideal I. If $I = 0$, then take $B = A$. Otherwise there is a nonconstant $f \in I$ (since otherwise I is the whole polynomial ring and $A = 0$). By the previous lemma, there is a subalgebra $B' \subseteq A' = k[x_1, \ldots, x_n]/(f)$ such that $B' \cong k[x'_1, \ldots, x'_{n-1}]$ and A' is finite over B'.

Let $I' = I/(f) \subseteq A'$ and $J = I' \cap B'$ be ideals. Then $B'/J \cong k[x'_1, \ldots, x'_{n-1}]/J$ is a finite subalgebra over $A'/I' = A$. By induction on the number of generators, the claim holds for B'/J, so there is a subalgebra $B \subseteq B'/J \subseteq A$ such that B is a polynomial ring $k[y_1, \ldots, y_m]$ and B'/J is a finite B-algebra. Since being a finite algebra is transitive, A is a finite B-algebra, as desired. ■

The Noether normalization lemma then implies the following result, which can be seen to be a sort of generalization of Nullstellensatz.

2.1.15 Proposition

Let L/k be a field extension such that L is a fg k-algebra. Then L is a finite extension of k (i.e. it is a finite k-algebra).

Proof

By Noether, L contains a polynomial subalgebra over which it is finite, and thus integral. Let $B = k[x_1, \ldots, x_n] \subseteq L$ be this algebra. Since L is integral over B, B must be a field (if $A \subseteq B$ is integral, A is a field iff B is a field), and so n must be zero. Thus $k \subseteq L$ is finite, as desired. ■

2.1.16 Corollary (Weak Nullstellensatz)

Let $A \neq 0$ be a fg algebra over an algebraically closed field k. Then for any maximal ideal $\mathfrak{m} \leq A$, $A/\mathfrak{m} \cong k$ as k-algebras.

Proof

Let $L = A/\mathfrak{m}$, which is a fg k-algebra, so by the above proposition it must be a finite. Algebraically closed fields admit no proper extensions, so $A/\mathfrak{m} \cong k$. ■

2.1.17 Corollary

Let $A \neq 0$ be a fg algebra over an algebraically closed field k. Then there exists a k-algebra homomorphism $\psi\colon A \to k$.

Proof

Choose a maximal ideal $\mathfrak{m} \leq A$, and consider $\psi\colon A \to A/\mathfrak{m} \cong k$. ■

Recall that the preimage of a prime ideal is always prime, but the preimage of a maximal ideal need not be maximal. For example under the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$, consider the preimage of the maximal ideal $(0) \leq \mathbb{Q}$, which is $(0) \leq \mathbb{Z}$; not maximal.

2.1.18 Proposition

Let A, B be fg k-algebras, and $f\colon A \to B$ a k-algebra morphism. Then for any maximal ideal $\mathfrak{m} \leq B$, $\mathfrak{m}_A = f^{-1}\mathfrak{m} \leq A$ is maximal.

Proof

The map f induces an injective k-algebra morphism $A/\mathfrak{m}_A \hookrightarrow B/\mathfrak{m}$. The latter is a field and a fg k-algebra, hence a finite k-algebra and in particular integral. Thus $A/\mathfrak{m}_A$ is an integral k-algebra as well, and since it is a domain ($\mathfrak{m}_A$ is prime), it must then be a field. So $\mathfrak{m}_A$ is maximal. ■

2.1.19 Definition

A ring A is **Jacobson** when $\mathrm{Jac}(A) = \mathrm{nil}(A)$.

In general we have $\mathrm{nil}(A) \subseteq \mathrm{Jac}(A)$, since $\mathrm{nil}(A)$ is the intersection of all prime and $\mathrm{Jac}(A)$ all maximal ideals.

2.1.20 Proposition

Every fg algebra over a field k is Jacobson.

Proof

Let A be a fg algebra over k, and $f \notin \mathrm{nil}(A)$. We then claim A has a maximal ideal not containing f. Since $f \notin \mathrm{nil}(A)$, $A_f \neq 0$ as the localizing set does not contain zero, and thus contains a maximal ideal $\mathfrak{m}$. Recall that $A_f \cong A[x]/(fx-1)$, so A_f is a fg k-algebra. Let $\iota\colon A \to A_f$ be the canonical map, then $\iota^{-1}\mathfrak{m}$ is maximal in A. Since $\iota f \in A_f^\times$, f is not in $\iota^{-1}\mathfrak{m}$ as desired. ■

2.1.21 Corollary

If A is a fg algebra over a field k, then for every ideal $I \leq A$,

$$\sqrt{I} = \mathrm{Jac}_A(I),$$

where $\mathrm{Jac}_A(I)$ is the Jacobson radical of I, the intersection of all maximal ideals containing I. In particular for a prime $\mathfrak{p} \leq A$,

$$\mathfrak{p} = \bigcap_{\mathfrak{p} \leq \mathfrak{m} < A} \mathfrak{m}.$$

Proof

This amounts to saying that A/I is a Jacobson ring, since pulling back $\mathrm{nil}(A/I)$ and $\mathrm{Jac}(A/I)$ give $\sqrt{I}, \mathrm{Jac}_A(I)$ respectively. Since A is a fg algebra over k, so is A/I and the result follows the above proposition. ■

Note:
Some authors define a Jacobson ring to be one where $\sqrt{I} = \mathrm{Jac}_A(I)$ for all ideals, or $\mathfrak{p} = \mathrm{Jac}_A(\mathfrak{p})$ for all prime ideals (these are equivalent). Note that this definition is more useful, but not how it was defined in this course. The above corollary shows that a fg algebra over a field is Jacobson in this regard as well.

Note that trhere is a natural bijection between $\mathbb{A}^n$ and k-algebra morphisms $\varphi\colon k[\mathbb{A}^n] \to k$.

(1) For every $P = (a_1, \ldots, a_n) \in \mathbb{A}^n$ define φ by $\varphi(x_i) = a_i$. That is, φ is the evaluation map ev_P given by $\mathrm{ev}_P(f) = f(P)$. In particular,
$$\ker \mathrm{ev}_P = \mathcal{I}(P).$$

(2) Conversely for $\varphi\colon k[\mathbb{A}^n] \to k$, let $P = (a_1, \ldots, a_n)$ by $a_i = \varphi(x_i)$. Then $\varphi = \mathrm{ev}_P$, so these are inverses.

Let $J \leq k[\mathbb{A}^n]$ be an ideal, then for every $P \in \mathbb{A}^n$, we have $P \in Z(J)$ iff $J \leq \mathcal{I}(P) = \ker \mathrm{ev}_P$ iff ev_P quotients over J: it induces $\overline{\mathrm{ev}}_P\colon k[\mathbb{A}^n]/J \to k$ where $\overline{\mathrm{ev}}_P(f + J) = f(P)$.

2.1.22 Proposition

For $P = (a_1, \ldots, a_n) \in \mathbb{A}^n$ we have
$$\ker \mathrm{ev}_P = (x_1 - a_1, \ldots, x_n - a_n),$$
which is a maximal ideal.

Proof

Since $\mathrm{ev}_P(x_i - a_i) = 0$ we have inclusion $\supseteq$. Since $k[\mathbb{A}^n]/\ker \mathrm{ev}_P \cong \mathrm{im}\,\mathrm{ev}_P \cong k$, $\ker \mathrm{ev}_P$ is maximal, it is sufficient to show $k[\mathbb{A}^n]/(x_1 - a_1, \ldots, x_n - a_n) \cong k$. Equivalently, the natural map $\iota\colon k \to k[\mathbb{A}^n]/(x_1 - a_1, \ldots, x_n - a_n)$ is surjective.
The projection $k[\mathbb{A}^n] \to k[\mathbb{A}^n]/(x_1 - a_1, \ldots, x_n - a_n)$ maps x_i to $\overline{x}_i = \iota(a_i)$. Thus ι must be surjective as the projection is. ∎

The following is another version of the Nullstellensatz:

2.1.23 Proposition

Let k be algebraically closed. Then the maximal ideals of $k[\mathbb{A}^n]$ are precisely the ideals $\ker \mathrm{ev}_P$ for $P \in \mathbb{A}^n$.

Proof

We saw that $\ker \mathrm{ev}_P$ are maximal, so let $\mathfrak{m}$ be maximal. By the weak Nullstellensatz, $k[\mathbb{A}^n]/k \cong k$, which gives a projection $f\colon k[\mathbb{A}^n] \to k[\mathbb{A}^n]/\mathfrak{m} \cong k$ with $\ker f = \mathfrak{m}$. But k-algebra morphisms $k[\mathbb{A}^n] \to k$ are all of the form ev_P for some P, so $f = \mathrm{ev}_P$ and thus $\mathfrak{m} = \ker \mathrm{ev}_P$. ∎

Now we can finally prove Nullsellensatz itself.

Proof (Nullstellensatz)

We already showed $\sqrt{J} \leq \mathcal{I}(Z(J))$, we now show the other direction. Let $f \in \mathcal{I}(Z(J))$, and consider $A = k[\mathbb{A}^n]/J$. By the above proposition, the maximal ideals of A are the ideals of the form $\ker \mathrm{ev}_P$ containing J, so $\ker \mathrm{ev}_P$ with $P \in Z(J)$. Because $f \in \mathcal{I}(Z(J))$, f vanishes at every point $P \in Z(J)$, so $f \in \ker \mathrm{ev}_P$ for every $P \in Z(J)$. Thus f is in every maximal ideal of A, so $\overline{f} \in \mathrm{Jac}(A)$. A is a fg k-algebra it is Jacobson, so $\overline{f} \in \mathrm{Jac}(A) = \mathrm{nil}(A) = \sqrt{J}/J$, thus $f \in \sqrt{J}$ as desired. ∎

Let us summarize the following formulations of Nullstellensatz:

2.1.24 Theorem (Equivalent formulations of Nullstellensatz)

The following are all equivalent formulations of Nullstellensatz, and all are true.

(1) Nullstellensatz: $\mathcal{I}(Z(J)) = \sqrt{J}$,

(2) If $J \neq (1)$ then $Z(J) \neq \emptyset$,

(3) Weak Nullstellensatz: if A is fg, $A/\mathfrak{m} \cong k$,

(4) If $A \neq 0$ is fg, then there is a $\varphi \colon A \to k$,

(5) Every maximal ideal of $k[\mathbb{A}^n]$ of the form $(x_1 - a_1, \dots, x_n - a_n)$.

2.1.25 Definition

For a topological space X, let $|X|$ denote the subspace of closed points.

Recall that for an affine scheme $\mathrm{Spec}(A)$, a closed point is just a maximal ideal, so $|X| \cong \mathrm{Specm}(A)$.

2.1.26 Corollary

Let k be algebraically closed, and $I \leq k[\mathbb{A}^n]$ an ideal. Then there is a bijection between the closed points of $\mathrm{Spec}(k[\mathbb{A}^n]/I)$ and $Z(I)$:

$$|\mathrm{Spec}(k[\mathbb{A}^n]/I)| \cong Z(I).$$

Proof

The closed points of the spectrum of $k[\mathbb{A}^n]/I$ are in bijection with the maximal ideals of $k[\mathbb{A}^n]$ containing I. These are of the form $(x_1 - a_1, \dots, x_n - a_n)$, and if it contains I then for $f \in I$ we must have that $f(a_1, \dots, a_n) = 0$. So mapping $(x_1 - a_1, \dots, x_n - a_n)$ to $(a_1, \dots, a_n) \in Z(I)$ is well-defined and clearly injective. Furthermore it is clearly surjective. ■

2.1.27 Corollary

Let A, B be fg k-algebras for some field k, and let $\varphi \colon A \to B$ be a k-algebra morphism. Then $\varphi^* \colon \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ restricts to a morphism of closed points $|\varphi^*| \colon |\mathrm{Spec}(B)| \to |\mathrm{Spec}(A)|$.

Proof

This is just a restatement of the preimage of a maximal ideal being maximal. ■